Trending Tool
User Guide

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All pertinent state, regional, and local safety regulations must be observed when installing and using this product. For reasons of safety and to help ensure compliance with documented system data, only the manufacturer should perform repairs to components.

When devices are used for applications with technical safety requirements, the relevant instructions must be followed.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

Failure to observe this information can result in injury or equipment damage.

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Safety Information

Important Information

NOTICE

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a Danger safety label indicates that an electrical hazard exists, which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

⚠ DANGER

DANGER indicates an imminently hazardous situation which, if not avoided, **will result in** death or serious injury.

⚠ WARNING

WARNING indicates a potentially hazardous situation which, if not avoided, **can result in** death or serious injury.

⚠ CAUTION

CAUTION indicates a potentially hazardous situation which, if not avoided, **can result in** minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

PLEASE NOTE

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

About the Book

At a Glance

Document Scope

Trending Tool is a software tool that offers an intuitive interface designed to assist your production resources. It enables you to obtain an accurate representation of how your installation is operating. It also allows you to select relevant variables from those available in the system, and place them in your application with a simple click. Once the first configuration is created and saved, the first recording is available on your computer hard disk. The recording will enable you to detect any operating system failures, and calculate your level of performance or draw up a monitoring plan.

Validity Note

This document is valid for the Trending Tool when used with Unity Pro 8.0 or later.

The technical characteristics of the devices described in this document also appear online. To access this information online:

Step	Action
1	Go to the Schneider Electric home page *www.schneider-electric.com*.
2	In the **Search** box type the reference of a product or the name of a product range. ● Do not include blank spaces in the model number/product range. ● To get information on grouping similar modules, use asterisks (*).
3	If you entered a reference, go to the **Product datasheets** search results and click on the reference that interests you. If you entered the name of a product range, go to the **Product Ranges** search results and click on the product range that interests you.
4	If more than one reference appears in the **Products** search results, click on the reference that interests you.
5	Depending on the size of your screen, you may need to scroll down to see the data sheet.
6	To save or print a data sheet as a .pdf file, click **Download XXX product datasheet**.

The characteristics that are presented in this manual should be the same as those characteristics that appear online. In line with our policy of constant improvement, we may revise content over time to improve clarity and accuracy. If you see a difference between the manual and online information, use the online information as your reference.

Chapter 1
Introducing Trending Tool

⚠ CAUTION
WRONG DATA INTERPRETATION Do not use the Trending function to diagnose real time processes. **Failure to follow these instructions can result in injury or equipment damage.**

The information displayed in the Trending function can be different than the information about the current state of the process.

⚠ CAUTION
WRONG DATA INTERPRETATION Do not consider the information displayed in the Trending function to be the real state of the process. **Failure to follow these instructions can result in injury or equipment damage.**

Due to scan time the Trending function can miss pulses and stealth changes.

⚠ CAUTION
WRONG DATA INTERPRETATION Take into account the scan time of Trending function when interpreting data. **Failure to follow these instructions can result in injury or equipment damage.**

This chapter discusses the Trending Tool product, and presents the subjects that will help you better understand this software and use it more effectively.

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
Advantages of Trending Tool	11
Using the Help	12
The Trending Tool Interface	14

Advantages of Trending Tool

At a Glance

Trending Tool is a software tool designed to help you monitor your production resources on a daily basis.

Trending Tool enables you to obtain an accurate representation of how your installation is operating. You can select the relevant variables from those available in the system, and place them in your application with a simple click. Within a few minutes, your first configuration is created and you can view your activity. Once your work is saved, your first recording is available on your computer hard disk.

These recordings will enable you to detect operating system failures, calculate your performance, and draw up a monitoring plan.

Trending Tool offers an intuitive interface with an efficient design to assist you with your tasks.

Using the Help

At a Glance

The Trending Tool online help features step-by-step instructions as well as all the essential information you need to perform the necessary procedures. It also provides answers which are directly relevant to your queries, omitting unnecessary information. To run the online help, press **F1** or click on the **?** menu and move the mouse pointer onto **Contents**.

If you have a question about a specific aspect of Trending Tool, you can type key words in the Search tab of the online help. Trending Tool then displays the sections of the online help related to your query.

Context-Sensitive Help

Clicking on the **Help** button in dialog boxes in Trending Tool opens an online help page explaining how to use these boxes. Dialog boxes also feature a **?** button in the top left-hand corner of the title bar. This button enables you to obtain information on each of the controls contained in the dialog boxes.

To use this function:

Step	Action
1	Click on the **?** button in the top left-hand corner of the dialog box title bar. This causes the shape of the mouse pointer to change.
2	Click on one of the dialog box's controls in order to display a small context-sensitive help window related to the selected control.
3	Click anywhere on the interface to close the context-sensitive window.

Information Contained in the Help

The online help is the quickest way to help you carry out the tasks you regularly perform to adjust your equipment. It enables you to find and quickly refer to the information you need to accomplish your task.

The Trending Tool online help comprises the following sections:

Section 1: Connecting Trending Tool to your installation

Your first step will be to connect Trending Tool to your equipment. This section also contains the tools available in Trending Tool to instantly perform your first connection to an installation.

Section 2: Adjusting with Trending Tool

This section explains how to carry out automated installation adjustment. Once you have set up the connection to the equipment, you will learn how Trending Tool can help you to improve your process. The displays, in the form of trend diagrams and functional analyses, give you real-time information about your installation's activity. The impact of your adjustments can be monitored through statistical indicators and measuring tools. You will also learn how to save your recordings in order to analyze how your installation is operating offline.

Section 3: Managing your recordings

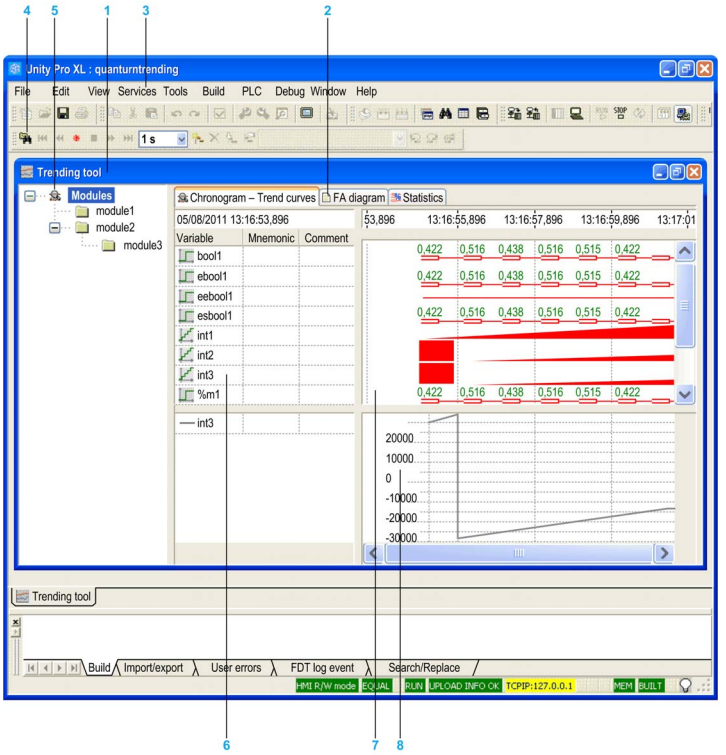
The third section presents the Trending Tool functions that enable you to achieve higher performance levels on your installation. By defining a reference cycle and using automatic divergence search, you can identify failures quickly. Calculated variables are performance indicators. They indicate which corrective actions need to be taken according to the production periods you have defined.

This section also contains information about how to print trend diagrams and use Trending Tool results in other applications.

The Trending Tool Interface

At a Glance

The Trending Tool interface complies with Microsoft Windows ergonomic standards. For more information on the common Microsoft Windows interface commands, refer to the user documentation for your operating system version.



1 Trending Tool displays recordings in standard Windows windows which can be moved, maximized or changed into icons within the application's main window.
2 FA diagram
3 The application commands are arranged by theme in drop-down menus.
4 The most common commands can be accessed via buttons on the tool bar and drop-down context-sensitive menus in the windows.
5 The configured modules tree

6 The panel containing statistical data
7 Chronogram
8 Trend Curve

Each time you execute a command (for example; start recording or delete variable), Trending Tool will display general message boxes. You can choose to disable these message boxes in the display dialog box *(see page 15)*.

Dialog Box Options

The performance related general message is enabled each time you open Trending Tool.

The **Dialog boxes** tab in the **Display options** dialog box enables you to show or hide a number of general message boxes.

Deleting variables	If this option is enabled, confirmation is required each time you attempt to delete a variable.
Start recording without saving the document	Select the radio button for the desired behavior. Before starting your recording, Trending Tool checks whether the file has been saved. If not, Trending Tool will: ● ask you if you want to save the file ● display the **save as** dialog box ● start recording without saving the trend (autosave will not work)
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Help	Open a window featuring an online help related to the dialog box properties.

Chapter 2
Connecting Trending Tool to Your Installation

What Is in This Chapter?

This chapter contains the following topics:

Topic	Page
Using PLC Items	18
Using PLC Timestamped Items	21
Using Modules	23
Data Acquisition	27
Saving of Recordings	28
Recording Options	29
Completing a Configuration	30

Using PLC Items

At a Glance

When you open the application, Trending Tool creates an empty document named **Trending Tool** by default. You can then add variables to the document to create a configuration.

Browsing PLC Variables

A PLC item corresponds to an automated system variable.

To browse for a variable, follow the steps below:

Step	Action
1	Click on the **Add variables..** command from the contextual menu in the Trend diagram tab. This command opens the Data Selection editor, which enables you to browse in the Data Selection dialog box.
2	If the item you are looking for does not appear, you can directly type its identifier in the editing field. **NOTE:** Special case for integer type items: an integer type item can correspond to an integer type variable but also to several boolean type variables. Thus, a 16-bit integer can correspond to 16 boolean type variables, each of them representing the developments of each of the bits of the item.

Adding Variables

Follow the steps below to add a variable:

Step	Action
1	Open the **Data Selection** dialog box (see above), enabling you to add a variable.
2	Browse the required items you want to monitor.
3	Depending on the type and number of selected items: **1.** Your selection contains one or more variables of the bit or floating type: in any case, you obtain boolean or floating variables. **2.** You have only selected one integer type item: you may create one integer variable or several boolean variables using the **Recording of an integer** dialog box.
4	Click on the **OK** button to open a dialog box featuring all the available choices described in step 3.
5	If an error message appears, check the syntax of the variable in the program. **NOTE:** Unknown type items are likely to be integers. Further information on properties of integers are in Recording of an Integer *(see page 20)*.

Renaming Variables

New variables are named according to the identifier of the item to which they are connected.

You can change this name at any time by following one of the procedures below:

Step	Action
1	Click twice (with a gap between clicks) on the name of the variable you want to change on the **Trending Tool variables** tree and change the name using the context-sensitive editor displayed by Trending Tool. Click on another area of the interface or press the **Return** key to close the editor and confirm your changes.
2	In the statistical data area of the recording (Trending Tool interface *(see page 14)*), click twice (with a gap between clicks) where the variable's name is displayed. The data table's cell changes into a text editor in which you can type the variable's name. Finally, press the **Return** key or click on another area of the interface to close the editor and confirm your changes.

The different properties of the variables existing in Trending Tool are detailed in variable properties *(see page 78)*.

Arranging variables in Trending Tool

The added variables are arranged in lists displayed by module. You can change the order of the variables at any time:

1. You can arrange the variables using the Copy & Paste and Drag & Drop functions in the statistical data area of the recording window (Trending Tool Interface *(see page 14)*). These functions enable you to easily move or copy variables from one module to the other.

You will find additional information on module management in the Using Modules *(see page 23)*.

Deleting Variables

To delete variables from a Trending Tool configuration, select the variables you want to delete in the statistical data area of the recording (Trending Tool Interface *(see page 14)*) and use the **Delete** command from the contextual menu or the Delete key.

Displaying the Properties of a Variable

Each Trending Tool variable has numerous properties *(see page 78)*.

To change these properties:

Step	Action
1	Select the variable in the recording window with the mouse pointer.
2	Use the **Properties..** command from the contextual menu to open the dialog box containing the variable's properties.

You can also use the context-sensitive menu (right button) in the statistical data area of the recording window to edit the properties of a variable, or click twice on the variable in rapid succession.

Recording Integers Dialog Box Options

The **Recording integers** dialog box enables you to select the acquisition mode for integer items transmitted to Trending Tool by the PLC.

The table below gives a brief explanation on each of the variables available on the dialog box:

As an Integer	Add the item to the current configuration as an integer. The variable will be a number represented by a series of values or a curve.
As a set of bits	Enables you to select only certain bits from those used by the item. Each of the selected bits will be represented in Trending Tool as a boolean value.
Is it?	This section enables you to specify the nature of the variables added to the configuration. For optimal readability, variables are differentiated by their nature in the Trending Tool FA diagram. You can change the defined nature of a variable at any time using the **Properties dialog** box of each variable.
An input	Select this option to specify that the added variables are automated system inputs or analyzed system inputs.
An output	Select this option to specify that the added variables are automated system outputs or analyzed system outputs.
An internal value	Select this option to specify that the added variables are values internal to the automated system or to the analyzed system. This is the default type given by Trending Tool to the variables.
Cancel	Cancel creation of the variables and close the dialog box.
Help	Open a window featuring an online help related to this dialog box.

Using PLC Timestamped Items

At a glance

It is possible to display in the Trending Tool a maximum of 16 variables scanned and timestamped at MAST scan time in the PLC. That allows to have the real state of the 16 variables at the PLC time scan. Other variables in Trending Tool are displayed with a time stamping set when the trending function receives the PLC notification.

Adding PLC Timestamped Variables

To add PLC timestamped items:

1. Check **Enable fast scanning for trending** check box in Variables leaf in Project Settings.
2. For each variable you want to scan by the PLC, check the **Scan Variable** property in **General attributes** tab in **Data Properties** window.
3. Rebuild and download project in the PLC (if necessary).

Then add PLC timestamped items using standard method : Adding PLC variables *(see page 18)*

Using PLC Timestamped Variables

NOTE: To use timestamped variables, the date and time of the PLC must be updated.

Fast scanned variables have the same behaviors than other Trending Tool variables. You can use fast scanned variable with standard variable but not mix them inside the same module.

Fast scanned variables are tag with an icon depending on its type plus  mark. The first variable added to trend module configures it like a fast or a standard module.

Fast modules can only own fast scan variables.Standard modules can only own standard variables.If user tries, an error will appear.

Using PLC Time for Reference

It is possible to choose the reference time to use for timestamping events. You can choose between PLC Time and PC Time.

If you choose PLC Time, Trending Tool reads Real-time clock of the PLC before starting the record and use it as reference time. In this case, check the PLC Real-time clock before recording.

If you choose PC Time, Trending Tool reads Real-time clock of the PC before starting the record and use it as reference time.

To choose reference time to use : Trend diagram options *(see page 47)*.

By default, PC Time is used as reference.

Limitations

Limitations are:

1. You cannot mix fast scanned variable with standard variable inside a same module.
2. You cannot fast scanned more than 16 variables.
3. This option is available only on EDT types (except STRING, DT, TOD, TIME and DATE).
4. This option is not available on Safety CPU, Premium & M340 CPU with version less than V2.9 and Quantum CPU with version less than V3.0.
5. The functionality was designed to be as fast as possible but fast scan get variables values only at MAST periodic time. So if a variable is modified more than one time per MAST cycle, you will only have its value at the end of MAST cycle.
6. If the Mast cycle is very short, it’s possible to lose some events. The quality is then considered like bad. In this case red hatching appear in Trending Tool time line.It is necessary to have a minimum duration cycle time upper than 5 ms. In case of CYCLIC period the cycle time is adjust by program.
7. When you mixed fast scanned variables with standard variables in Trending Tool, you can have poor performance on standard variables if the mast period is short.
8. The trending of fast scanned variables need to connect Unity Pro and PLC via USB or Ethernet link.
9. The trending of fast scanned variables is only available if the MAST task is configured as a periodic task, not as a cyclic task.

Using Modules

At a Glance

When you open the application, Trending Tool creates an empty document named Trending Tool by default. This document contains a first module named **Trending Tool module**.

To easily manage recordings, you can create a modules tree diagram in which the variables to monitor will be placed. For example, you can use modules to group together variables belonging to the same mechanical movement, or variables involved in a failure you have already encountered on your installation.

Trending Tool imposes no limits as regards to the number of variables or modules a configuration can contain. The same variable can be present at different places in a configuration.

Adding Modules to a Configuration

To add a module to your configuration:

Step	Action
1	Select the parent module of the module you want to create on the modules tree.
2	Use the **Insert module** command from the contextual menu. This command creates a sub-module related to the selected module and opens a context-sensitive text editor in the window.
3	Type the name of the module in the text editor and confirm your choice by pressing the **Return** key or clicking on another area of the interface.

NOTE: You can also use the context-sensitive menu by right-clicking on the parent module of the module you want to create.

Adding Variables to a Module

To add variables to a module, follow one of the procedures below:

Step	Action
1	From the module tree (Trending Tool Interface *(see page 14)*), select the module to which you wish to add variables.
2	Use the **Add variables...** command from the contextual menu to open the **Data Selection** dialog box, and create the variables *(see page 18)* using this dialog box.

You can also select variables from other modules of the same configuration by copying or moving them using the Copy & Paste function, as described below:

Step	Action
1	Using the Copy & Paste function, select the variables in a module and use the **Cut** command from the contextual menu if you wish to move these variables, or use the **Copy** command from the same menu if you wish to duplicate them.
2	Select the module to which you wish to add the variables by clicking on its name on the module tree. Use the **Paste** command from the contextual menu to place the variables in this new module.

The Drag & Drop function also enables you to move or copy variables in Trending Tool, as described below:

Step	Action
1	Select the variables you want to move or copy in the statistical data area of the recording window.
2	Hold your finger down on the left mouse button to move the selected variables to the destination module and release the button. If you wish to copy the variables, press the **Ctrl** key while you move the mouse.

Displaying the Contents of a Module

To immediately display the contents of a particular module of the configuration, click on the required module in the modules tree area of the recording window.

Moving a Module

Trending Tool also enables you to move modules defined in your file. Follow the procedure below:

Step	Action
1	Select the module you want to move on the module tree.
2	Hold your finger down on the left mouse button to move the selected module to the destination module.

NOTE: You can only move a module if the module in which it is located is different from the destination module.

Renaming a Module

New modules are always named New module by default. You can change this name at any time by following one of the procedures below:

Step	Action
1	Click twice (with a gap between clicks) on the name of the module you want to change on the module tree (see Trending Tool interface *(see page 14)*), and change the name using the context-sensitive editor displayed by Trending Tool. Click on another area of the interface or press the **Return** key to confirm your changes or **Esc** to cancel them.
2	Right-click on the name of the module you want to change on the module tree and select the **Rename** command from the context-sensitive menu. This command opens a context-sensitive text editor where the name of the module is located. Change the name written in the editor and click on another area of the interface or press **Return** to confirm your changes. Press **Esc** to cancel them. The modules are automatically sorted into alphabetical order by Trending Tool.

Displaying the Properties of a Module

Every Trending Tool module has numerous properties. *(see page 25)* To change these properties:

Step	Action
1	Select the module in the recording window with the mouse pointer.
2	Use the **Properties...** command from the contextual menu to open the dialog box containing the properties of the module.

NOTE: You can also use the context-sensitive menu (right button) on the module tree to edit the properties of the module.

Viewing the Description Tab Options of a Module

The **Description** tab details all the description parameters of the modules.

Dialog box options	Description
Name	Name of the module. To change the name of a module, use the Rename command from the context-sensitive menu of the modules tree.
Contents	This section details the number of variables and the number of modules descending from the module.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Viewing the Comment Tab Options of a Module

This tab provides information about the purpose of the module and the identity of its author.

The **Comment** tab provides information on the role of the modules.

Dialog box options	Description
Author	Type the name of the person who created the module in this editor.
Description	Type the description of the module's role or the reason for its creation in this editor.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Viewing the Display Tab Options of a Module

The **Display** tab allows you to modify the trend diagram line height.

Dialog box options	Description
Customize	Allows you to define the desired height and choose whether or not to display the scale of values.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Four predefined size values are available:

Size	Pixels	Scale Displayed
S	25	no
M	90	yes
L	175	yes
XL	350	yes

Deleting Modules

To delete a module from a Trending Tool configuration, select the module you want to delete on the module tree of the recording (see Trending Tool Interface *(see page 14)*) and use the **Delete** command from the contextual menu or the **Del** key. If the module you want to delete contains sub-modules or variables, Trending Tool will display a message asking you to confirm your action by clicking on **OK**.

Data Acquisition

At a Glance

As soon as the variables are configured, you can start acquiring data. Trending Tool connects to the PLC and starts the acquisition of developments.

The machine status changes can be viewed interactively on the screen using the two diagram modes below:

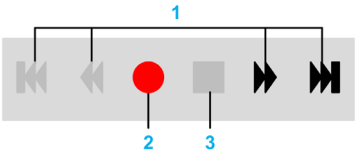
- Trend Diagram Mode *(see page 32)*
- Functional Analysis Mode *(see page 34)*

Start Data Acquisition

To start the data acquisition registration, click the **Recording** button  on the recording tool bar shown in the illustration below.

Stop Data Acquisition

To stop the data acquisition registration, click the **Stop** button  found on the recording tool bar, shown in the illustration below.



1 Displacement in the registration
2 Start the registration
3 Stop the registration

Saving of Recordings

Principals

When the user click Stop or when the set duration *(see page 29)* expires, the acquisition stops and the recording is saved.

Launching the acquisition for an unlimited duration may cause the following problems:

- The file can reach a size that would make it unusable
- Closing Trending tool stops the acquisition

When you start a new acquisition from a document already containing developments, Trending tool asks you if you want to keep these developments. If the answer is yes, the acquisition starts in a new document containing a copy of the initial structure of the document.

Autosave

When recording, you can use the autosave in order to limit the risks of losing information in case the system suddenly stops. This function automatically runs the **File/Save** command at regular intervals.

NOTE: The autosave is an option *(see page 29)* to activate before starting the acquisition as described.

Recording Options

Introduction

The conditions of saving and acquisition can be set.

Accessing the Options

To specify the saving conditions, select he Trending Tool window, then select **Services** → **Recording Options**.

The options are distributed into two tabs:

- General dedicated to saving
- Manual dedicated to limiting the acquisition time

NOTE: Recording options are not accessible during acquisition.

General Tab

The **General** tab contains various options:

- **Activate**: activates the autosave
- **Save period**: time between two saves when the autosave is activated
- **Warning!**: this function is activated for all documents, provided they are linked with a file.

Manual Tab

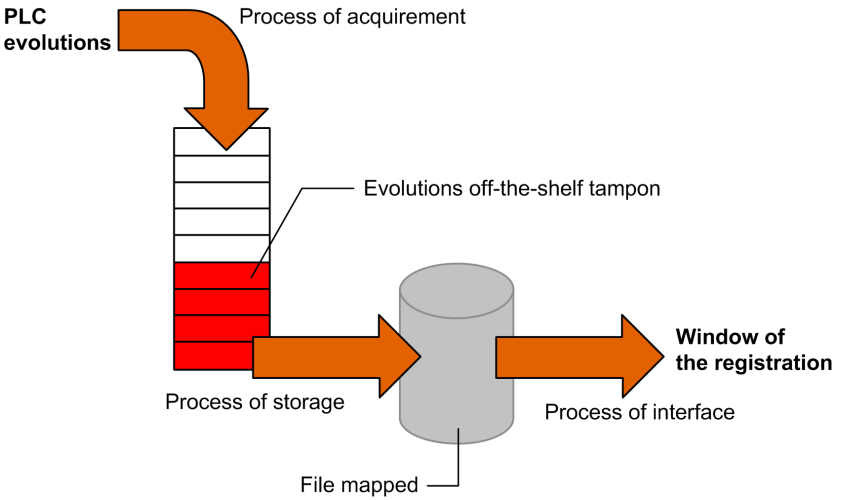
- **Infinite duration**: when checked, the recording time is disabled, therefore only a user can stop the recording.
- **Recording time**: desired recording duration

Completing a Configuration

Saving and Completing Simultaneously

In Trending Tool, you can only add variables in stop mode. All Trending Tool configuration functions are available before, during and after recordings. Find more information on the configuration change functions in Connecting Trending Tool to Your Installation *(see page 17)*.

The Trending Tool application is based on a multi-process architecture, so the accuracy of the recordings cannot be altered by user actions on the interface, as demonstrated by the diagram below. Queries from PLC are immediately released by a dedicated Trending Tool process for each server. Their only function is to temporarily store the transmitted developments in a memory area.



The developments stored in the temporary memory area are then archived in the recording file through another process. The interface is controlled by a supplementary process which accesses the recording file according to the needs of the interface.

Chapter 3
Viewing Changes to Variables

What Is in This Chapter?

This chapter contains the following topics:

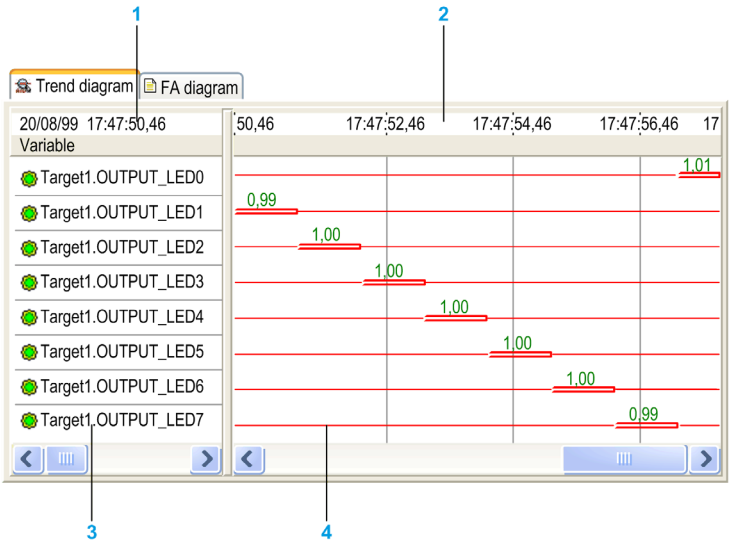
Topic	Page
Trend Diagram Mode	32
Functional Analysis Mode	34

Trend Diagram Mode

At a Glance

The trend diagram represents the sequence of the system's developments. The display is similar to that of a pen recorder. It can be accessed from the **Trend diagram** tab in every recording window.

Trend Diagram



1 Date and time: The date displayed on the left-hand side of the dating scale is that of the trend diagram's left edge. This date changes as you move on the diagram.

2 Dating scale: The dating scale enables you to date the different developments in the recording. You can place time stamps or select the mechanical mode display using this scale.

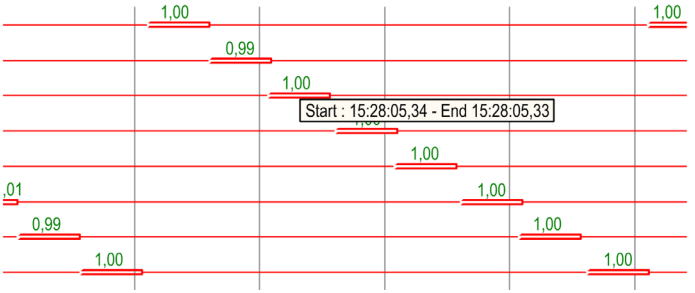
3 Statistical Table: The data table on the left-hand side of the trend diagram area provides information related to variable statistics.

4 Signal trend diagram: Each variable is displayed according to numerous display options. You can obtain additional information on displayed developments by moving the mouse pointer over them. You can also superimpose your recording with reference cycles saved beforehand.

Getting Times of Events

In addition to the display of the developments along with their duration, the Trending Tool trend diagram can easily give you the exact position of every development:

1. Move the mouse pointer onto one of the diagram's developments.
2. Wait for approximately 5 seconds without moving the mouse: Trending Tool will display a small pop-up window featuring the start time and end time of the pointed development.
3. Move the mouse to close this window.



Expanding Trend Diagram

You can expand the trend diagram line, by following the steps below, to help you use analogical variables:

1. Right click on the trend diagram to display the context-sensitive menu.
2. Click on **Select line height** and select the height required as described in display tab menu *(see page 26)*.

NOTE: There is also an optional display mode: Trend Curves *(see page 46)*.

All the trend diagram functions are described in the next two chapters of this help: Adjusting with Trending Tool *(see page 37)* and Working with your recordings *(see page 55)*.

Functional Analysis Mode

At a Glance

The functional analysis mode provides a detailed list of the developments of the process.

The Data Table

Each development is listed in a table in chronological order. Only the variables included in the selected module are listed. The scroll bar, on the right-hand side of the window, enables you to scroll through the table rapidly. The size of the scroll bar depends on the relative size of the data zone displayed in the window in comparison with the size of the whole recording.

For each development, the user can obtain the following information:

Comment	Comment on the variable that has changed status.
Start date	The date (day/month/year) on which the status change occurred.
End date	The date (day/month/year) on which the new status disappeared.
Duration	The duration of the development.
Start time	The time (hour:minutes:seconds,milliseconds) at which the new status appeared.
End time	The time (hour:minutes:seconds,milliseconds) at which the new status disappeared.
Mnemonic	The mnemonic of the variable that has changed status.
Value	The new value of the variable that has changed status.
Variable	The name of the variable that has changed status.

It may be difficult to calculate values based on two successive developments of the variable, simply because the next development has not occurred yet. In such a case, the value indicated by Trending Tool is replaced by a single hyphen. There are different ways to configure the data table. As specified in this section, you can add *(see page 34)*, change *(see page 35)* or delete *(see page 36)* columns in order to modify the Trending Tool default display. The table can be copied *(see page 100)* to other office IT applications. You can also change the display of the data lines.

Adding a Column to a Table

In the FA Diagram mode, proceed as follows to add a column to the data table:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to insert a column and right-click.
2	Use the **Insert column** command from the context-sensitive menu. This command opens a new column dialog box *(see page 35)*.
3	Complete the dialog box with the information you wish to obtain and click on **OK**.

Viewing FA Display Options

This dialog box enables you to manage the display of the FA diagram columns or statistical data table in trend diagram mode.

Dialog box options	Description
Item	Select the type of information you need in the drop-down list.
Title	Type the title of the column in this editor. If the field is left blank, Trending Tool will choose a default title depending on the item you selected.
Alignment	Select the alignment mode for the data in the column from the three options in the drop-down list. The data can be left or right aligned or centered in the column.
Width	Set the column's width in this field, either by directly typing a number in the editor or by changing the displayed width using the arrows to the right of the editor.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Changing Information in a Column

In the FA Diagram mode, proceed as follows to change the data contained in a table column:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to change a column and right-click.
2	Use the **Properties...** command from the context-sensitive menu. This command opens a column properties dialog box *(see page 35)*.
3	Change the dialog box according to the information you wish to obtain and click **OK**.

Moving a Column

Proceed with the following steps, using the Drag & Drop method, to reorganize columns in a table:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to move a column and left-click.
2	Hold your finger down while moving the mouse. The column is gray toned and superimposed over the banner.
3	Once the column has been moved to the required place, release the mouse button.

Deleting a Column

In the FA Diagram mode, proceed as follows to delete a column from the data table:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to delete a column and right-click.
2	Use the **Delete column** command from the context-sensitive menu.

Changing the Display

There are three different types of Trending Tool variable: inputs, outputs and variables internal to the control system. To easily analyze the operation of an automated system, you may control the action and reaction sequences. With Trending Tool, you can allocate a color to each type of variable and therefore view the action sequences easily.

Proceed with the following steps to differentiate the lines in the data table:

Step	Action
1	Move the mouse pointer onto the table and right-click.
2	Select the **Display...** command from the context-sensitive menu. This command opens a dialog box defining the FA Diagram display parameters *(see page 36)*.
3	Set up the display parameters in the dialog box and click on **OK**.

Viewing FA Display Mode

This dialog box enables you to configure the display of the FA diagram mode. The display depends on the types of variable configured in Trending Tool. By default, all the variables are considered as internal variables in Trending Tool.

Dialog box options	Description
Background Color	This area determines the display options for the different lines of the data table.
Inputs	From the drop-down menu, select the display color for the developments of the variables defined as being command system inputs.
Outputs	From the drop-down menu, select the display color for the developments of the variables defined as being command system outputs.
Internal variables	From the drop-down menu, select the display color for the developments of the variables defined as being command system internal variables.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Chapter 4
Adjusting with Trending Tool

Overview

This chapter details how to adjust automated installations. It explains how to recover and analyze the information acquired directly through the connection to the system, during and after the recording.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
4.1	Using Records	38
4.2	Measuring Differences and Detecting Synchronisms	49
4.3	Activity Statistics	52

Section 4.1
Using Records

At a Glance

When recording or analyzing developments, you can use the various Trending Tool functions to give you a better understanding of the contents of a recording. These functions are detailed below.

What Is in This Section?

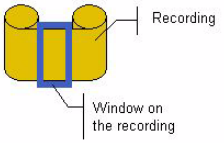
This section contains the following topics:

Topic	Page
Scrolling Through a Recording	39
Search Function	41
Changing the Scale of a Diagram	43
Mechanical Mode or Standard Mode	44
Arranging Variables	45
Using Trend Curves	46

Scrolling Through a Recording

Scroll Bars

The windows in Trending Tool only display part of the recordings, see diagram below. The scroll bars enable you to scroll through a recording quickly and efficiently.



1 Recording
2 Recording window

In trend diagram mode, the scroll bar is located at the bottom of the diagram. The size of the scroll bar depends on the relative width of the window in comparison with the size of the recording. For example, if you change the scale of the diagram *(see page 43)*, the size of the scroll bar will vary. You can scroll through the recording just like any other office IT application. The position reached is constantly indicated in the top left-hand corner of the trend diagram window.

In functional analysis mode, the scroll bar is located on the right-hand side of the window. The width of the scroll bar depends on the number of developments displayed in the window in comparison with the total number of developments contained in the recording.

Switching Between the Two Modes

The **Trend diagram** and **FA diagram** tabs enable you to switch between the two diagram modes quickly. When you scroll in one mode, this is also applied in the other mode. For example, you can use trend diagram mode to reach a broad zone of the recording and fine-tune your position in the FA diagram mode.

The Recorder Tool Bar

Scrolling through diagrams is made even easier with the Trending Tool Recorder tool bar, which works just like the control panel of a video recorder.

You can use the following buttons to perform these functions:

Function	Description
Beginning	Displays the beginning of the recording.
Previous page	Displays the previous page of the recording.
Next page	Displays the next page of the recording.
End	Display the last page of the recording.

The other buttons on this tool bar enable you to start *(see page 27)* or stop *(see page 27)* the recording of developments.

Finding a Particular Event

Trending Tool enables you to find particular events in a recording, as listed below:

- Search for the next rising or falling edge of a boolean variable
- Search for the next status change of a variable
- Search for a time stamp

To perform a search in a recording, proceed as follows:

1. Use the contextual menu to open the Trending Tool Find dialog box *(see page 41)*.
2. Select the various criteria in the dialog box and click on the **OK** button to display the particular event in the recording window.

NOTE: You can also use the shortcut button  on the **Standard** tool bar to open the **Find** dialog box.

Search Function

At a Glance

The **Find** dialog box allows you to find variables, events and time stamps in a Trending Tool document. It therefore helps you to find a location in the modules, variables and trend diagram.

Three kinds of objects can be searched for:

1. Variables
2. Events
3. Time stamps

Search for Variables

This function allows you to find variables in the selected module. A variable is found if the key word is contained in its name, its mnemonic or its comment.

Selecting a variable in the output list selects the module and the variable in the list of variables.

Searching for Events

This function allows you to find the status changes of a variable and its statuses by length.

Selecting an event in the results list centers the trend diagram around this event and the variable concerned is selected. Double-clicking on an event adds a time stamp on the date of the event in the module where the variable is located.

Search criteria depends on the variable type.

Boolean Variable	
Rising Edge	Search rising edge or value equal to true.
Falling Edge	Search falling edge or value equal to false.
Minimum Activity time and Maximum Activity Time	Allow you to search for the beginning of variable activity statuses (value = TRUE) by restricting the activity length (in h:mn:s).
Minimum Inactivity Time and Maximum Inactivity Time	Allow you to search for the beginning of variable activity statuses (value = TRUE) by restricting the activity length (in h:mn:s).

Integer and Floating Variable	
Minimum threshold	Allows you to search for dates during the value of variable that are below the threshold.
Maximum threshold	Allows you to search for dates during the value of variable that are above the threshold.

Searching for Time Stamps

This function displays the time stamps present in the selected module.

Selecting a time stamp in the results list will cause the trend diagram to be centered around the time stamp.

Changing the Scale of a Diagram

Available Scales

Several scales can be applied to the trend diagram. These are:

Scale	Recorded time per displayed centimeter
10 mn/cm	10 recorded minutes per displayed centimeter
1 mn/cm	1 recorded minute per displayed centimeter
0.1 mn/cm	6 recorded seconds per displayed centimeter
0.01 mn/cm	600 recorded milliseconds per displayed centimeter
0.001 mn/cm	60 recorded milliseconds per displayed centimeter
30 s/cm	30 recorded seconds per displayed centimeter
10 s/cm	10 recorded seconds per displayed centimeter
1 s/cm	1 recorded second per displayed centimeter
500 ms/cm	500 recorded milliseconds per displayed centimeter
100 ms/cm	100 recorded milliseconds per displayed centimeter
50 ms/cm	50 recorded milliseconds per displayed centimeter

The displayed centimeter is based on the data transmitted by the operating system. These data may not reflect the exact reality.

Changing the Scale

To change the scale, use the **Zoom in/out** commands from the contextual menu.

You can also use the **Scale** drop-down list on the **Standard** tool bar to apply a scale change.

Mechanical Mode or Standard Mode

At a Glance

Trending Tool processes dates in standard format (hours : minutes : seconds , milliseconds) or in mechanical format (hours : minutes : hundredths of minutes).

To switch between the two modes at any time, proceed as follows:

Step	Action
1	Use the **Mechanical mode** command from the contextual menu. An annotation is displayed by this command when the mechanical mode is active.
2	Move the mouse pointer onto the trend diagram's dating scale and right-click. This causes a context-sensitive menu to open. Use the **Mechanical mode** command from this menu to switch between the two display modes.
3	Press **F2** to switch between the two modes instantly.

Arranging Variables

At a Glance

You can arrange the variables of a module by following the order in which rising edges appear. This function is very useful as it gives you an instant overview of the sequence of an action/reaction cycle within the analyzed system.

Proceed as follows:

Step	Action
1	Move the mouse pointer immediately in front of the first rising edge of the cycle you want to arrange.
2	Right-click to open the context-sensitive menu and use the **Arrange variables** command from this menu. This command arranges the list of the module's variables according to the order in which the rising edges of the variables appear.

Using Trend Curves

At a Glance

The **Trend Curves** view is a fitted representation of analogical variable. It allows comparing different curves with a precise scale. In addition, the **Probe Mode** displays value of each variable at a specific date.

Displaying the Variable Trend Curve

To display the **variable trend curve**, proceed as follows:

Step	Action
1	Select the variable in the list on left hand side of the chronogram.
2	Select Display Trend Curve in the context menu, by right clicking on the variable.

If the Chronogram view is displayed in full screen, this action will automatically split the screen to show at the same time Chronogram and Trend Curves.

Hiding the Variable Trend Curve

To hide the variable trend curve, proceed as follows:

Step	Action
1	Repeat previous actions to display the **Trend Curve**.
2	Select the variable in the list on left hand side of the **Trend Curves**.
3	Select Hide the variable in the context menu.

If after this action, no trend curve is displayed, then the Trend Curves view is automatically hidden.

Navigating Between Chronogram and Trend Curves

Trending Tool allows navigation between Chronogram view and Trend Curves view.

You can execute the following actions, by selecting the **Display** menu, in the view context menu of Chronogram or Trend Curves:

Step	Action
1	**Hide**: Hide the current view.
2	**Screen Split**: This action is available when a view is in full screen. It allows displaying simultaneous the **Chronogram** and the **Trend Curves** view. The occupancy ratio by default between the two views can be modified in the display dialog box. *(see page 47)*
3	**Maximize**: This action is only available when the window is split. It shows in full screen the current view.

Viewing Diagram Options

The **Trend diagram** tab in the **Display options** dialog box enables you to modify the display options of the trend diagram. The dialog box options are listed below in the table.

Dialog box options	Description
Use PLC Time for reference	If this option is enabled, reference time for events is PLC time based and not PC time based.
Display in milliseconds	If this option is enabled, durations are displayed to an accuracy of 1ms instead of 10ms. This option doesn't improve the accuracy of the information coming from PLC at the time of the acquisition.
Optimize the durations display	On the trend diagram, this option allows you to display only status durations that are compatible with the display scale rather than all durations.
Trend diagram view area	In Trending Tool, the split settings between the trend diagram and trend curves are saved in the module. This option allows you to customize Trending Tool. You can set up these options: **Value for new module:** define the size of the trend diagram in the window (as a percentage) when viewing a new module. **Value for Screen Split action:** define the size of the trend diagram in the window (as a percentage) after this action
Background color	You can customize display options by selecting background colors for the trend diagram and trend curves.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Help	Open a window featuring an online help related to the dialog box properties.

Using Curve Graphical Representation

By default, module variables have different representations. If a variable is present in different modules, then, its representation may be different in each module. The representation is able to be customized with the properties dialog box.

The Properties dialog box shows Trending Tool variable characteristics. It contains two tabs: **Description** and **Display**.

The **Display** tab allows you to customize the graphical representation of trend curves. The different properties of the Display dialog box are listed below:

Type	Two choices are available: a standard representation (two points linked by a line) or a stepped representation.
Color	Curve color.
Line size	Curve thickness.
Style	Hachure style.
Point	Type of point used to show the measurement points.

The Display scale tab dialog box variable characteristics are listed below:

Display Scale	
Display the scale	Show or hide the variable scale.
Upper Bound	Enable maximum scale value setting if manual is checked, otherwise the maximum statistic value is used.
Maximum statistic value	Show the maximum value that the variable is able to take for the duration of the recording.
Lower Bound	Enable minimum scale value setting if manual is checked, otherwise the minimum statistic value is used.
Minimum statistic value	Show the minimum value that the variable is able to take for the duration of the recording.

Modifying the Scale

To show or hide the variable scale, proceed as follows:

Step	Action
1	Select the variable in the list on left hand side of the **Trend Curves**.
2	Select **Show the scale** in the context menu, by right clicking on the variable.

The lower and upper bound can be modified on the properties dialog box *(see page 47)*.

Horizontal marks displayed behind trend curves correspond to the most right of displayed scale.

You can change the order of the scales by doing a drag and drop.

All actions are available by right clicking on the variable on the left hand side of trend curves. They are also available by right clicking on the scale.

Using the Probe Mode

The **Probe Mode** allows you to display values for a precise date. Thus, the values, located at the intersection between probe and curve, are shown in the display zone of the scale.

To show or hide the probe, proceed as follows:

1	Click right on the trend curves.
2	Select the Probe Mode item in the context menu.

To move the probe, proceed as follows:

1	Move the mouse cursor on the probe
2	Press and maintain the mouse left button and move it

Section 4.2
Measuring Differences and Detecting Synchronisms

Accessing Time Stamps

Displaying a Time Stamp

To display a time stamp, proceed as follows:

Step	Action
1	Move the mouse pointer onto the trend diagram's dating scale where you wish to insert a time stamp.
2	Right-click to open a context-sensitive menu.
3	Use the Insert time stamp command from that menu. This command places a time stamp on the trend diagram. If you left-click on the dating scale, a time stamp will immediately be positioned on the trend diagram. **NOTE:** During the recording, you cannot place time stamps on the **moving** area of the trend diagram's window. This applies to all time stamp management functions in Trending Tool. Time stamps are only added to the printing file if the **Print time stamps** box is checked in the **Options** tab of the document page setup *(see page 93)* properties dialog box.

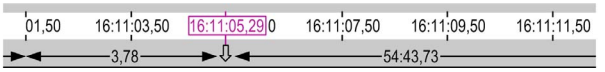
Modifying a Time Stamp Attribute

You can differentiate time stamps by modifying their color and/or by displaying the date corresponding to their position on the trend diagram's dating scale.

Use the dating scale's context-sensitive menu and proceed as follows:

Step	Action
1	Move the mouse pointer onto the arrow of the time stamp you wish to change.
2	Right-click to open the context-sensitive menu.
3	Use the **Time stamp properties** command and then one of the related subcommands to change one of the time stamp's attributes.

When the **Show position** attribute is selected, the date corresponding to the position of the time stamp is placed on the dating scale, as shown below. If this attribute is active, the corresponding item (**Show position**) from the dating scale's context-sensitive menu is checked. The **Color** context-sensitive submenu enables you to choose one of the 15 colors available for the time stamp. The active color is checked in this submenu.



Moving a Time Stamp

By moving the mouse pointer onto one of the dating scale's time stamps and by holding your finger down on the left mouse button, you can move the time stamp. Once the required position is reached, release the left button. If, while you are moving the time stamp, you approach one of the trend diagram's edges, Trending Tool will automatically shift the diagram in the required direction.

You can move and create a time stamp in one step.

Synchronizing a Time Stamp With the Edge of a Variable

When you create or move a time stamp, you can specify the edge of the variable with which it must be lined up. To do this, move the mouse pointer onto the time stamp on the trend diagram's dating scale and hold your finger down on the left mouse button. Then, move the mouse until you reach the required edge of the trend diagram. Release the left button. If the time stamp is hooked to the edge of the variable, the edge will be surrounded by a square in Trending Tool. If other trend diagram edges are synchronous with the selected edge, they will also be surrounded by a square.

This synchronization mode works with the rising and falling edges of boolean variables and with the value changes of integer and floating variables.

Deleting a Time Stamp

To delete a time stamp, proceed as follows:

Step	Action
1	Move the mouse pointer onto the trend diagram's dating scale where you want to delete a time stamp.
2	Right-click to open the context-sensitive menu.
3	Use the **Delete** command from this menu.

If the mouse pointer is not moved onto the time stamp, the **Delete** command will be gray-toned.

Locating the Position of a Time Stamp

To locate the position of a time stamp, move the mouse pointer onto the time stamp in question and wait for approximately five seconds. Trending Tool will display a small pop-up window featuring the dates and times related to this time stamp. To close this window, simply move the mouse.

Measuring a Difference

To measure a difference between two positions, proceed as follows:

Step	Action
1	Create the first time stamp and place it at the beginning of the area you want to measure.
2	Create the second time stamp and place it at the end of the area you want to measure.
3	You can apply a scale change *(see page 43)* to view both time stamps at the same time. Trending Tool displays the distance between both time stamps on the trend diagram's dating scale.

Section 4.3
Activity Statistics

Indicators Available in Trending Tool

At a Glance

The data table in the statistics tab allows you to obtain various statistics about the activity of the process. These statistics are related to the entire recording. During recording, the data in the table are refreshed every second.

Viewing the Data Table Variables

Each variable of the module is listed in the data table. The scroll bar on the right-hand side of the window enables you to scroll through the table. The scroll bar at the bottom of the table enables you to view the table's right-hand columns. The size of the scroll bars depends on the relative size of the area displayed in the window in comparison with the total size of the table.

The information below is available for each variable:

Comment	The comment linked to the variable.
Total activity time	The total activity time of a boolean variable.
Total inactivity time	The total inactivity time of a boolean variable.
Maximum activity time	The longest activity period of a boolean variable.
Maximum inactivity time	The longest inactivity period of a boolean variable.
Minimum activity time	The shortest activity period of a boolean variable.
Minimum inactivity time	The shortest inactivity period of a boolean variable.
Average activity time	The average time of every activity of a boolean variable.
Average inactivity time	The average time of every period of inactivity of a boolean variable.
Average status time	The average time during which an integer or a floating variable maintains a value.
Activity frequency	The frequency at which the rising edges of a boolean variable appear.
Inactivity frequency	The frequency at which the falling edges of a boolean variable appear.
Mnemonic	The mnemonic of the variable. In a configuration, the mnemonics are automatically initialized by the application if the latter provides the function of the variable, or can be added using the Neutral file... command *(see page 30)* from the **File** menu.
Number of rises	The number of times a boolean variable reaches 1.

Number of changes	The number of value changes of an integer or floating variable.
Number of falls	The number of times a boolean variable reaches 0.
Line number	The number of a line in the table.
Average value	The mean value of an integer or floating value.
Variable	The name of the variable.

Calculations that do not apply to a particular type of variable (the number of activations of an integer variable or the average value of a boolean variable) are represented by empty cells in the table.

Different ways exist to configure the data table. You can add, change and delete columns in order to modify the standard Trending Tool display. You can also copy *(see page 100)* the table to other office IT applications.

Adding a Column

Proceed as follows to add a column to the data table:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to insert a column and right-click.
2	Use the **Insert column** command from the context-sensitive menu. This command opens a new column dialog box *(see page 35)*.
3	Complete the dialog box with the relevant information and click on **OK**.

Changing a Column

Proceed as follows to change the data contained in one of the table's columns:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to change a column and right-click.
2	Use the **Properties...** command from the context-sensitive menu. This command opens a column properties dialog box *(see page 35)*.
3	Change the configuration of the dialog box according to the relevant information and click on **OK**.

You can change the size of a column directly by moving the mouse pointer onto the left or right edge of that column. The shape of the pointer then changes into a vertical bar with two horizontal arrows. To change the size of the column, left-click and hold your finger down. Move the mouse to adjust the width of the column. Once the required width is obtained, release the button. The width of the column is updated instantly.

Moving a Column

You can drag and drop columns to change the organization of the table. Proceed as follows:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to move a column and left-click.
2	Hold your finger down to move the column using the mouse. The column is gray toned and superimposed over the banner.
3	Release the mouse button.

Deleting a Column

Proceed as follows to delete a data table column:

Step	Action
1	Move the mouse pointer onto the column header banner where you want to delete a column and right-click.
2	Use the **Delete** column command from the context-sensitive menu.

Chapter 5
Working With Your Recordings

At a Glance

Trending Tool offers other functions for handling the recordings.

What Is in This Chapter?

This chapter contains the following sections:

Section	Topic	Page
5.1	Comparing with a Reference Cycle	56
5.2	Calculated Variables in Trending Tool	65
5.3	Elaborate Statistics on Boolean Variables	85
5.4	Printing Your Recordings	90
5.5	Exchanging Data With Another Application	100
5.6	Managing your Recordings	101

Section 5.1
Comparing with a Reference Cycle

At a Glance

The reference cycles defined in Trending Tool are used to compare the recordings with development sequences saved beforehand.

What Is in This Section?

This section contains the following topics:

Topic	Page
Managing Reference Cycles	57
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Managing Reference Cycles

At a Glance

Reference cycles enable you to store periods that characterize how an installation works. These periods are stored in Trending Tool and can be used in all subsequent recordings containing the module.

Viewing a Reference Dialog Box

This dialog box determines the active and inactive variables at the beginning of a reference cycle. These parameters are used to place a reference cycle in a recording.

Dialog Box Options	Description
Variables of the Module	Click on the variables in the list that you want to add to or remove from the selection. The variables selected from the list are the variables that will be tested at the beginning of a reference cycle.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Help	Open a window featuring an online help related to the dialog box properties.

Viewing Reference Cycle Options

The Properties dialog box displays the characteristics of a Trending Tool reference cycle.

The Description tab in the Properties of a reference cycle dialog box displays the general characteristics of a Trending Tool cycle.

Dialog box options	Description
Settings	This area of the dialog box determines the internal characteristics of a reference cycle.
Name	Type the name of the reference cycle in this area.
Tolerance(s)	Type the difference analysis triggering threshold in this area. The value must be in seconds and between 0.1 and 60 seconds.
Display	This area details the display options for the reference cycle.
Color	Select a display color from the drop-down list.
Activity time display	Check this box to display the duration of each development of the reference in the trend diagram.
Position	This area details the position of the reference cycle in the recording.
Beginning	Date and time of the beginning of the reference cycle.
End	Date and time of the end of the reference cycle.
Duration	Duration of the reference cycle.
OK	Close the dialog box and apply the changes made.

Dialog box options	Description
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

The Confirmation tab in the Properties of a reference cycle dialog box displays the equation associated with the start of a reference cycle.

This tab enables you to view and change the criteria of the beginning of the cycle.

Dialog box options	Description
Starting event	The starting event is the rising or falling edge selected in the trend diagram as the starting signal for the reference cycle.
Active Variables at Beginning of Cycle	This area contains the list of the variables that must be active at the beginning of the cycle.
Change...	This button opens a dialog box for selecting the variables that must be active/inactive at the beginning of the cycle.
Inactive Variables at Beginning of Cycle	This area contains a list of the variables that must be inactive at the beginning of the cycle.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying any changes.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Creating a Reference Cycle

To create a reference cycle, proceed as follows:

Step	Action
1	Display the module concerned in the trend diagram mode and use the **Define a reference cycle...** command from the **Insert** menu. This command automatically places a reference cycle in the module. By default, the beginning and end of cycle time stamps are placed at 20% and 80% of the displayed zone respectively.
2	Move the beginning and end of cycle time stamps until you reach the required positions. Reference cycle time stamps are moved just like Trending Tool standard time stamps *(see page 49)*.

Step	Action
3	Adjust the properties of the cycle with the **Reference cycle properties...** command from the context-sensitive menu (right-click) of the trend diagram area. This command opens the reference cycle properties dialog box. **NOTE:** If the module already contains a reference cycle, Trending Tool opens a dialog box asking the following question: “This module already contains a reference cycle. Do you want to add a second one to this module?”. Select **Yes** to add a second cycle, **No** to replace the previous reference cycle with a new one, or **Cancel** to cancel the creation of a new reference cycle.

Creating a Triggering Condition

In Trending Tool, the start of a reference cycle can be governed by a particular condition comprising:

Step	Action
1	The rising edge or falling edge of a variable.
2	The list of the module’s variables that must be active at the start of the reference cycle.
3	The list of the module’s variables that must be inactive at the start of the reference cycle.

In Trending Tool, you must specify a triggering condition on a reference cycle to carry out automatic divergence searches. To add triggering conditions to a reference cycle:

Step	Action
1	Display the reference cycle on which you want to work.
2	Move the mouse pointer onto the beginning of cycle time stamp and hold your finger down on the left mouse button to move this time stamp.
3	Move the mouse until you reach the edge of the variable that will be used as the beginning of cycle condition. Release the left button to hook the beginning of cycle time stamp to this event. The selected edge is surrounded by a square in Trending Tool.
4	Use the **Reference cycle properties...** command from the trend diagram’s context-sensitive menu. To open this context-sensitive menu, move the mouse pointer onto the trend diagram’s window and right-click. This command opens a dialog box *(see page 57)* detailing the reference cycle properties.
5	When the reference cycle is synchronized with a variable edge by the user, Trending Tool introduces the following default condition: the module’s boolean variables must have a status identical to the one it had before the user selected the edge. You can click on the **Validation** tab in the dialog box to change this configuration.

Step	Action
6	Click on the **Change...** button in the **Active variables at beginning of cycle** area to add or remove variables that must be active at the start of the cycle. This causes a dialog box *(see page 60)* to open, showing a list of the module's variables. Click on the items in the list to select or deselect them. Confirm the changes made by clicking on the **OK** button in the dialog box.
7	Proceed in the same manner to draw up the list of variables that must be inactive at the start of the reference cycle.

Showing a Reference Cycle

To show a reference cycle:

Step	Action
1	Select the trend diagram mode.
2	Use the **Reference cycle** command from the View menu. This command displays the module's first reference cycle.
3	If the module contains several reference cycles, select the required cycle from the drop-down list on the Reference cycle tool bar.
4	If the reference cycle does not appear, use the **Show reference cycle** command from the **Recording** menu.

You can also use the **Show reference** button  on the standard Trending Tool tool bar or press Alt+F2 to display the reference cycle. This button is held down as long as a reference cycle is displayed.

The Show reference command can be accessed via the **Show reference** button  on the Trending Tool reference cycle tool bar.

Hiding the Reference Cycle

To hide the reference cycle:

Step	Action
1	Use the **Reference cycle** command from the **View** menu or the **Show reference** button  on the standard tool bar or press Alt+F2.

Changing a Reference Cycle

Trending Tool associates several properties which are all detailed in a dialog box with each reference cycle.

To open this dialog box and change the properties of a reference cycle:

Step	Action
1	Display the required reference cycle on the trend diagram.
2	Use the **Reference cycle properties...** command from the trend diagram's context-sensitive menu. To open this context-sensitive menu, move the mouse pointer onto the trend diagram's window and right-click. This command opens a dialog box *(see page 57)* detailing the properties of the reference cycle.
3	Change the properties as required in the dialog box and apply the changes made by clicking on **OK**.

Accessing a Reference Cycle Property

Reference cycles are linked to modules. You can also access the reference cycle properties from the properties of the modules. Proceed as follows:

Step	Action
1	Select the module containing the reference cycle on the module tree in the recording window.
2	Use the **Properties...** command from the **Edit** menu. This command opens a dialog box detailing the properties of the selected module.
3	Click on the **Reference cycle** tab. This tab contains a list of the reference cycles defined for this module.
4	Select the required reference cycle from the list and click on the **Properties...** button to open the dialog box detailing the properties of the reference cycle.
5	Change the properties as required in the dialog box and apply the changes performed by clicking on **OK**.

Deleting a Reference Cycle

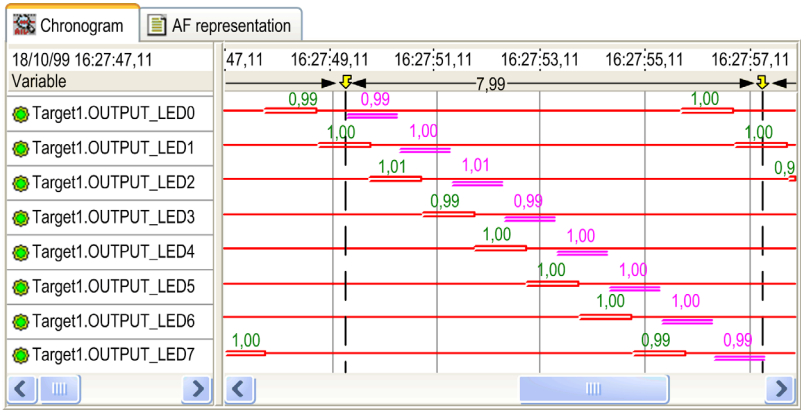
To delete a reference cycle:

Step	Action
1	Select the module containing the reference cycle on the module tree in the recording window.
2	Use the **Properties...** command from the **Edit** menu. This command opens a dialog box detailing the properties of the selected module.
3	Click on the **Reference cycle** tab. This tab contains a list of the reference cycles defined for this module.
4	Select the required reference cycle from the list and click on the **Delete** button to delete the reference cycle.

Juxtaposing the Reference Cycle and the Trend Diagram

At a Glance

You can use reference cycles to compare developments recorded at a given moment with previously saved developments. In Trending Tool, you can superimpose the reference cycle and the recording to spot differences between them. The reference cycle juxtaposition functions only work for reference cycles with a starting event *(see page 57)*. The **Reference cycle** tool bar contain these functions.



Moving the Reference Cycle

You can use the mouse to move the reference cycle in the recording. Proceed as follows:

Step	Action
1	Move the mouse pointer onto one of the developments of the reference cycle and hold your finger down on the left mouse button.
2	Keep the button pressed down until you reach the required location for the reference cycle. Trending Tool automatically shifts the recording when you approach a border of the window.
3	Release the left button. The reference cycle is now placed as required.

In Trending Tool, there is also a specific function that enables you to move the reference cycle directly to any place in the recording:

Step	Action
1	Display the recording as required and use the Show reference cycle command from the Recording menu. This command places the reference cycle that you are currently viewing at the center of the trend diagram. You can also use the Show reference cycle button  on the Reference cycle tool bar.

Switching Between the Cycles of a Recording

If the reference cycle has a triggering condition, it can be automatically juxtaposed with all the cycles of a recording using the Previous reference cycle and Next reference cycle commands from the Recording menu. These commands shift the position of the reference cycle forward and backward a cycle respectively on the trend diagram. The display of the trend diagram is also adjusted accordingly.

These two actions can be performed via the Previous  and Next  buttons on the Reference cycle tool bar.

Comparing the Durations of Events

The reference development display is one of the properties of each reference cycle. You can display these durations by following the procedure described in the paragraph entitled Changing a reference cycle *(see page 57)*.

Moreover, you can obtain information on each development of a reference cycle by moving the mouse pointer directly onto the required development.

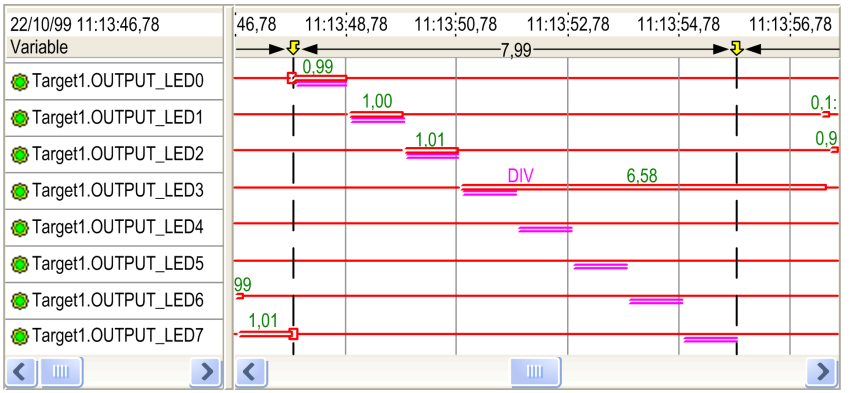
Proceed as follows:

Step	Action
1	Move the mouse pointer onto one of the developments of a reference cycle. After five seconds, Trending Tool will open a pop-up window detailing the characteristics of the development. The information provided concerns the current position of the development compared with its position at the beginning of the cycle and the duration of the development.
2	Move the mouse pointer to close the pop-up window.

Detecting Divergence

At a Glance

Trending Tool can compare an entire recording with a previously saved reference cycle and automatically recover the divergences it contains. The diverging event is indicated on the reference cycle by the term DIV, as indicated on the diagram below.



Divergences appear when the difference between the edge of a variable in the reference cycle and in the recording is above the threshold defined in the reference cycle properties *(see page 57)*. By default, the difference is set at 200 milliseconds.

To find a divergence:

1. Display the beginning of the recording, or the moment preceding the possible position of the divergence at the very least. Trending Tool will only search for a divergence in the section following the displayed part of the recording.
2. Use the Find divergences command from the Recording menu to start searching for a divergence. Trending Tool will stop as soon as it finds a divergence or at the end of the recording if no divergence can been found.

Section 5.2
Calculated Variables in Trending Tool

At a Glance

Calculated and FlipFlop variables in Trending Tool help users to adjust automated installations. They enable you to detect drifts or measure saturations and queues. This section will further discuss topics on calculated variables.

What Is in This Section?

This section contains the following topics:

Topic	Page
Back to Performance Indicators	66
Defining Calculated Variables	69
Defining FlipFlop Variables	75
Counters	77
Variable Properties	78
Predefined Calculated Variables	81
Calculated variable options	83
Constant definitions	84

Back to Performance Indicators

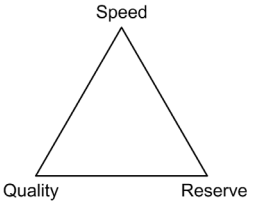
This section contains general information about the assessment of an automated installation. You will find an example of the distribution of operating times (known as status times) as well as topics covering how to create performance indicators.

Measuring Performance

The performance of an installation or piece of equipment depends on three specific factors: trend, quality, and operational availability.

- **Trend** is the production speed of the machine
- **Quality** is the proportion of good parts in the total number of produced parts
- **Availability** is the effective production time in comparison with the machine's start-up time

These factors must be compared with the nominal values agreed with the machine's manufacturer to assess the installation's overall output.



Status Times

The total running time of an automated installation can be broken down into a succession of periods called statuses. Analyzing the status times of an installation provides useful information about how the installation is running, and well though-out and efficient solutions can therefore be applied to improve the systems. There are several possible breakdowns of the status times by jobs and experiences. Below is a breakdown based on the CNOMO E41150520N standard.

Total time	Required time	Running	Good running	Good parts production
				Bad parts production
				Others
			Corrupt running	Good parts production
				Bad parts production
				Others
		Proper stop	Failure	Undetected failure
				Awaiting maintenance
				Diagnostic
				Tool supply
				Spare parts supply
				Repair = corrective maintenance action
				Return to service
				Others
			Functional stop	Scheduled tool change
				Manufacturing change
				Control
				Regular servicing
				Others
		Induced stop		Saturation
				Lack of parts
				Lack of operator
				Lack of energy
				Non-standard upstream parts
				Others
	Non-required time			Break
				Systematic servicing
				Conditional servicing
				Unused
				Others

This table below can also be explained using the following definitions:

Total time	Reference period covering all the possible resource statuses (24 hours, week, month, etc.
Required time	Time during which the user applies resources for production purposes.
Running time	Time during which the resources perform the function in accordance with the specifications.
Downtime	Period of time during which the resources cannot perform their function following a failure (random part of the proper stop time). A failure is a fault in an element of the resources preventing the latter from performing their function.
Functional stop time	The functional stop time covers the scheduled part of the proper stop time (scheduled tool change time, manufacturing change time, regular servicing time, etc.)
Induced stop time	Period of required time during which the resources cannot perform their function for external reasons.
Proper stop time	Period of required time during which the resources cannot perform their function for a reason related to the resources themselves.

The Role of Trending Tool

By analyzing the signals recorded by Trending Tool on your installation, you will be able to set up calculated variables that represent the different status times of your installation. The statistical data table linked to each of these variables (whether the signal comes directly from the machine or whether it is a composite signal calculated by Trending Tool) provides all the relevant information such as the activation time of each variable and the average activation time. For example, if you set up an indicator representing the running time, Trending Tool immediately calculates the average time for each run and the total running time for the recording period. You can precisely analyze this indicator by breaking down the data by production periods that represent a shift or a manufacturing batch. Moreover, comparing seasonal recordings (one per month, one per week, etc.) will also enable you to monitor your installation's development trend and measure your installation's growth in productivity.

In Trending Tool, the calculated variable concept is the basis of all the performance measurement actions you will be able to perform. The following chapters feature additional information on Trending Tool functions:

- Defining calculated variables *(see page 69)*: presentation of how to create calculated variables in Trending Tool.
- Activity statistics: *(see page 52)* presentation of the different indicators calculated in Trending Tool for each of the variables.

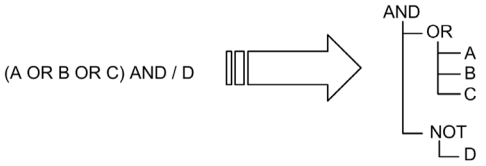
Defining Calculated Variables

At a Glance

You can create calculated boolean, integer, and floating variables. There are also wizards *(see page 81)* to help you to create variables with pre-formatted equations (Extract a bit as an integer, create an integer with a boolean variable and convert a BCD to an integer).

The Equation Tree

The equations of the variables calculated in Trending Tool are represented as a structured tree. The coding principle used is similar to that of the Polish code (the way that HP calculators work). The example below illustrates the technique used to reproduce parenthesis levels in Trending Tool.



Available Operators

Operators		Operands	Results	Description
Boolean operators	**AND**	from 1 to N boolean terms	Boolean	Performs the logical **AND** between all the terms.
	OR			Performs the logical **OR** between all the terms.
	NOT	1 boolean term		Returns the negation of the term.
	XOR	2 boolean terms		Returns TRUE if the two terms are different.
	DELAY	1 boolean term		Delays the activation or deactivation of the term.
Comparison operators	**EQUAL TO**	2 integer or floating terms		Returns TRUE if both terms are equal, or else FALSE.
	GREATER THAN			Returns TRUE if the first term is strictly superior to the second one, or else FALSE.
	GREATER THAN OR EQUAL TO			Returns TRUE if the first term is superior or equal to the second one, or else FALSE.
	LESS THAN			Returns TRUE if the first term is strictly inferior to the second one, or else FALSE.
	LESS THAN OR EQUAL TO			Returns TRUE if the first term is inferior or equal to the second one, or else FALSE.
	DIFFERENT FROM			Returns TRUE if both terms are different, or else FALSE.

Operators		Operands	Results	Description
Arithmetical operators	**PLUS**	2 integer or floating terms	Integer	Returns the sum of the values of both terms.
	MINUS			Returns the subtraction of the values of both terms.
	MULTIPLY			Returns the multiplication of the values of both terms.
	DIVIDE			Returns the result of the division of the first term by the second one. If the second term is equal to zero, the result arbitrarily returned by Trending Tool will be zero.
	QUOTIENT			Returns the quotient of the division of the first term by the second one.
	REMAINDER			Returns the remainder of the division of the first term by the second one.
	CONSTANT	-		Returns the value of the constant.
	CONSTANT TIME	-		Returns the number of second from midnight to selected time as a constant.
	CURRENT TIME	-		Returns the number of second from midnight to chronogram current time as a constant.
	SQUARE	1 integer or floating term	Integer or floating	Returns the value of the term to the square.
	SQUARE ROOT			Returns the square root of the value of the term.
	EXPONENTIAL			Returns exp(value of the term).
	LOGARITHM			Returns log(value of the term).
	NAPIERIAN LOGARITHM			Returns ln(value of the term).
	SINE			Returns sin(value of the term).
	COSINE			Returns cos(value of the term).
	TANGENT			Returns tan(value of the term).

Operators		Operands	Results	Description
Logical operators	**LOGICAL AND**	2 integer terms	Integer	Returns the result of the **AND** between the two integer terms.
	LOGICAL OR			Returns the result of the **OR** between the two integer terms.
	LOGICAL XOR			Returns the result of the **XOR** between the two integer terms.
	LOGICAL NOT	1 integer term		Returns the complement of the integer term.
Conversion operators	**INTEGER->BOOLEAN**		Boolean	Returns FALSE if the value of the term is 0, TRUE otherwise.
	BOOLEAN-> INTEGER	1 boolean term	Integer	Returns 1 if the value of the term is TRUE, 0 if the value of the term is FALSE.
Specific operators	**TIMER**	1 boolean term	Integer	Shows the activity length of a boolean variable in milliseconds. The value changes on the falling edge of the boolean variable.

Viewing the Calculated Variable Dialog Box Options

The new calculated variable dialog box enables you to create an equation to determine how the calculated variable will operate.

Dialog Box Tabs	Description
Equation	Trending Tool boolean equations are represented as a tree. This area of the dialog box shows the equation of the calculated variable.
Operators	This tab enables you to add operators to the equations of calculated variables. Available operators are represented by the leaves of the tree shown in the tab. To add an operator, select it on the tree and click on the Add button on the left-hand side. ● Add: This button enables you to add the selected operator to the equation currently being created. ● Remove: This button enables you to remove the selected operator from the equation. ● Show all operators: This option is used to hide the least common operators.

Dialog Box Tabs	Description
Variables	The following two headings appear in this tab: **Modules** and **Trending Tool variables**. ● Modules: This drop-down list contains all the modules present in the configuration. ● Trending Tool variables: This list contains the variables of the selected module. ● Add: This button enables you to add the selected operator to the equation currently being created. ● Remove: This button enables you to remove the selected operator from the equation.
OK	This buttons confirms the changes made to the equation and closes the dialog box.
Cancel	This button closes the dialog box without applying the changes made to the variable's equation.

Creating a Calculated Variable

Calculated variables can be created in Trending Tool at any time, just like any other variable, even during or after a recording.

To create a calculated variable in a module, proceed as follows:

Step	Action
1	On the module tree, select the module to which you wish to add a calculated variable.
2	In the Insert/Calculated variable menu, choose the type of variable to create. This command opens the New calculated variable dialog box *(see page 83)*.
3	In the dialog box, select the root operator of the equation in the list of the **Operators**, click then on the button to Add. In Trending Tool equations, the root operator is always an operator that returns a result of the chosen type.
4	Select a first term for the root operator, either from the list of **Variables** to add a variable directly to the equation, or from the list of the Operators to add a sub-equation, then click the button again to Add.
5	Repeat steps 3 and 4 until a comprehensive equation is obtained and click on the **OK** button to add the variable to the module and close the dialog box. The **OK** button is only active if the syntax of the equation is correct.

Changing a Calculated Variable

Calculated variables are boolean variables. They have the same display and analysis properties. In addition, the calculated variable properties dialog box *(see page 78)* features an Equation tab *(see page 74)*.

To change the equation of a calculated variable:

Step	Action
1	Select the variable you want to change in the statistical table of the variables.
2	Use the **Properties...** command from the **Edit** menu. This command opens the dialog box containing the properties of the calculated variable.
3	Click on the **Equation** tab in the dialog box. The **Equation** tab shows the variable's current equation.
4	Click on the **Change...** button to open the equation editor.
5	Change the variable's equation as required and click on the **OK** button to save your changes and close the editor. The **Cancel** button enables you to cancel the changes and close the editor at the same time.
6	Click on the **OK** button to close the calculated variable properties dialog box. The developments of the variable are immediately recalculated for the whole of the recording.

Viewing the Equation Tab for a Variable

The **Equation tab** displays the equation of a calculated variable.

Dialog Box Options	Description
Equation	This area enables you to view the equation associated with the variable in the form of a tree.
Change	Click on this button to open the dialog box to change the variable's equation.
OK	Close the dialog box and apply the changes made.
Cancel	Close the dialog box without applying the changes made.
Apply	Apply the changes made without closing the dialog box.
Help	Open a window featuring an online help related to the dialog box properties.

Deleting a Variable Linked To a Calculated Variable

Calculated variables are linked to the variables used in their equations. If you try to delete these variables, Trending Tool will display an alert message preventing you from deleting them.

Defining FlipFlop Variables

FlipFlop Variable Operating Principle

In Trending Tool, a FlipFlop variable is a memory which changes status according to condition-related events.

An Trending Tool FlipFlop variable is characterized by four attributes:

Reach 1	Reach 0
Starting event (mandatory)	Ending event (optional)
Starting condition (optional)	Ending condition (optional)

Definition of an event and of a condition:

- An event is the combination between a boolean variable and a (rising or falling) edge.
- A condition is a boolean equation.

Behavior of a FlipFlop variable:

- It reaches 1 if the starting event has the expected edge and the starting condition is true.
- It reaches 0 if the ending event has the expected edge and the ending condition is true.

Restriction:

A Trending Tool FlipFlop variable cannot directly or indirectly depend on itself.

Viewing the FlipFlop Variable Dialog Box

The **New FlipFlop variable** dialog box enables you to configure the parameters of a FlipFlop variable.

Dialog Box Controls	Description
Starting or Ending Event	Defining an event: ● the selected button enables you to choose a variable of the configuration. ● the  and  buttons indicate which edge (rising or falling) of the selected variable will generate events.
Starting Condition or Ending Condition	A condition is a calculated variable *(see page 65)* contained in the FlipFlop variable. Edit: this button enables you to create or change a condition. Delete: this button deletes the existing condition.
OK	This button confirms the changes made to the equation and closes the dialog box.
Cancel	This button closes the dialog box without applying the changes made to the variable's equation.

Creating a FlipFlop Variable

You can create FlipFlop variables at any time in Trending Tool, just like with any other variable, even during or after a recording.

To create a FlipFlop variable in a module, proceed as follows:

Step	Action
1	Display the module tree to select the module to which you wish to add a calculated variable.
2	Use the **FlipFlop variable...** command from the Insert menu. This command opens the new FlipFlop variable dialog box *(see page 75)*.
3	In the dialog box, select a starting event and an ending event (boolean variables).
4	Starting or ending conditions are optional as an empty condition is always true.

Changing a FlipFlop Variable

FlipFlop variables are boolean variables. They have the same display and analysis properties. The FlipFlop variable properties dialog box *(see page 78)* features an Equation tab.

To change the equation of a FlipFlop variable:

Step	Action
1	Select the variable you wish to change in the variable statistical table.
2	Use the **Properties...** command from the **Edit** menu. This command opens the dialog box displaying the properties of the FlipFlop variable.
3	Click on the **Equation** tab in the dialog box in order to change the current FlipFlop variable.
4	Click on the **Change...** button to open the equation editor.
5	Customize the FlipFlop variable as required and click on the **OK** button to save your changes and close the editor. The **Cancel** button enables you to cancel any changes and close the editor at the same time.
6	Click on the **OK** button to close the FlipFlop variable properties dialog box. The developments of the variable are immediately recalculated for the whole of the recording.

Deleting a Variable Linked To a FlipFlop Variable

The FlipFlop variables are linked to the variables acting in their events and conditions. If you try to delete such variables, Trending Tool will display an alert message preventing you from deleting them.

Counters

How Counters Function

In Trending Tool, a counter is a calculated integer variable with memory. Its value is changed by specific operations.

There are three different kinds of operation:

- Add
- Subtract
- Allocate

An operation therefore involves adding, subtracting or allocating a value when the associated event occurs and its condition is true.

If the events of several operations occur simultaneously, then operations are run in the order that the operations are created.

A minimum and maximum value can be defined for a counter variable.

Variable Properties

At a Glance

The **Properties** dialog box displays the characteristics of a Trending Tool variable. The tabs found in the box depend on the type of variable.

Description Dialog Box Options

The **Description** tab details all the description parameters for the Boolean, Integer; Floating, and Calculated variables.

Variable Type	Variable Options	Description
Boolean Integer Floating Calculated	Server	Name of the server to which the variable is connected. This field stays empty if the variables are calculated variables.
	Item	Name of the item as it was defined for the connection to the server. This field also stays empty if the variables are calculated variables.
	Name	Name of the variable. You can change this name by typing a new one in the editor.
	Nature	Selects the nature of the variable from the three options in the drop-down list. Input corresponds to variables of the automated system entry type or analyzed command system entry type. Output corresponds to automated system outputs or analyzed command system outputs. Internal variable is the type allocated by default to variables by Trending Tool.
	Mnemonic	The variable's mnemonic. The mnemonic is a short text describing the variable. You can change the mnemonic by typing new text in the relevant field in the editor.
	Comment	The comment on the variable. The comment is a long text describing the variable's role. You can change the comment by typing new text in the relevant field in the editor.
	OK	Closes the dialog box and apply the changes made.
	Cancel	Closes the dialog box without applying any changes.
	Apply	Applies the changes made without closing the dialog box.
	Help	Opens a window featuring an online help related to the dialog box properties.

Display Dialog Box Options

The Display tab defines the display properties of the variables on the trend diagrams for the Boolean, Integer; Floating, and Calculated variables.

Variable Type	Variable Options	Description
Boolean Integer Floating Calculated	Display format	This section specifies the type of diagram to apply to the numerical variable.
	Simple curve	Represents the developments of the variable as a thin line.
	Area	Represents every development of the variable as a full rectangle.
	Series of values	Represents every development of the variable as a horizontal line with the value displayed above it.
	Colors	This section enables you to configure the display colors for the simple curves and areas in the trend diagram mode.
	Normal	Selects the display color for the curve or area when the value is between the minimum and maximum values.
	Exception	Selects the display color for the curve or area when the value is outside the range defined for the minimum and maximum values.
	Scale	This area enables you to set minimum and maximum values for the display of the curves and areas.
	Minimum value	Check the box by the heading to manually set a minimum display value. Once the box is checked, the text editor to the right becomes active and enables you to type a new value. In Trending Tool, the default minimum value is the lowest value in the whole of the recording.
	Maximum value	Check the box by the heading to manually set a maximum display value. The Maximum value field is similar to the Minimum value field.
	OK	Closes the dialog box and applies the changes made.
	Cancel	Closes the dialog box without applying any changes.
	Apply	Applies the changes made without closing the dialog box.
	Help	Opens a window featuring an online help related to the dialog box properties.

Equation Dialog Box Options

The equation tab defines the properties on the trend diagrams for calculated variables.

Variable Type	Variable Options	Description
Calculated	Equation	This area enables you to view the equation associated with the variable in the form of a tree.
	Change	Click on this button to open the dialog box to change the variable's equation.
	OK	Closes the dialog box and applies the changes made.
	Cancel	Closes the dialog box without applying any changes.
	Apply	Applies the changes made without closing the dialog box.
	Help	Opens a window featuring an online help related to the dialog box properties.

Predefined Calculated Variables

Building-up an Integer Variable

This function allows you to build one integer variable from several boolean variables using the Building up an integer variable from boolean variables dialog box.

Using the Boolean Variable Dialog Box

The **Create an integer variable from boolean variables** dialog box creates a calculated integer variable *(see page 65)*, where the value of the bits corresponds to the value of other boolean variables. The equation model used is:

VAR1*1 + VAR2*2 + VAR3*4 + VAR4*8 + VAR5*16 + VAR6*32 + VAR7*64 + VAR8*128 + ...

VARx is a boolean variable.

Dialog Box Options	Description
Define the Integer Variable Size	Number of bits to edit.
Integer Variable	List of the value of all bits to edit.
Boolean Value	Three mark values are available: ● Constant 0 and 1 ● Status of an existing boolean variable: Modules: This drop-down list features all the modules available in the configuration. Trending Tool variables: This list contains all the variables of the selected module.
OK	This button allows you to create the calculated variable described above and closes the dialog box.
Cancel	This button closes the dialog box.

Extraction of Boolean Variable(s)

Constraints	The selected variables must be integer variables.
Results	This function allows you to extract bits from one integer variable. So, boolean variables are created using the Extraction of boolean variable(s) from an integer variable dialog box.

Using Extract Boolean Variable(s) Dialog Box

The Extract boolean variable(s) from an integer variable dialog box creates calculated boolean variables that represent the status of the bits of an integer variable. The equation model used for boolean variables is:

INTEGER->BOOLEAN((INTEGER VARIABLE) LOGICAL AND (CONSTANT = BIT NUMBER))

Dialog Box Options	Description
Integer variable	Contains the selected integer variable from which you want to extract the boolean variables.
Select the bits you want to record	If you chose to follow some bits of the variable only, you must click the buttons in this zone to determine precisely which bits to follow.
OK	This button allows you to create the calculated variables described above and closes the dialog box.
Cancel	This button closes the dialog box.
Help	Open a window featuring an online help related to the dialog box.

Convert BCD Into Integer

Constraints	The selected variables must be integer variables.
Results	For each selected variable, a new calculated variable *(see page 65)* is created which value is equal to BCD variable value. The template of the equation is: ((VARIABLE) LOGICAL AND (CONSTANT = 0x0F)) PLUS ((((VARIABLE) DIVIDE (CONSTANT = 0x0010)) LOGICAL AND (CONSTANT = 0x0F)) MULTIPLY (10)) PLUS ((((VARIABLE) DIVIDE (CONSTANT = 0x0100)) LOGICAL AND (CONSTANT = 0x0F)) MULTIPLY (100)) PLUS ((((VARIABLE) DIVIDE (CONSTANT = 0x1000)) LOGICAL AND (CONSTANT = 0x0F)) MULTIPLY (100)) PLUS where VARIABLE is supposed to be a BCD type variable.

Calculated variable options

At a Glance

The **New** calculated variables in the **Recording options** dialog box enables you to create an equation to determine how the calculated variable will operate.

Equation

The Trending Tool boolean equations are represented as a tree. This area of the dialog box shows the equation of the calculated variable.

Operators tab

The Operators tab enables you to add operators to the equations of calculated variables. The available operators are represented by the leaves of the tree shown in the tab. To add an operator, select it on the tree and click on the **Add** button on the left-hand side.

Controls:

- **Add**: this button enables you to add the selected operator to the equation currently being created
- **Remove**: this button enables you to remove the selected operator from the equation
- **Show all operators**: this option opens a window featuring an online help related to the dialog box properties

Variables tab

Lists:

- **Modules**: drop-down list that contains all the modules present in the configuration
- **Trending Tool variables**: list that contains the variables of the selected module

Buttons:

- **Add**: this button enables you to add the selected variable to the equation currently being created
- **Remove**: this button enables you to remove the selected variable from the equation

Close buttons

Buttons:

- **OK**: this button confirms the changes made to the equation and closes the dialog box
- **Cancel**: this button closes the dialog box without applying the changes made to the variable's equation

Constant definitions

At a Glance

The **Constant value** dialog box enables you to associate a value with a constant used in an equation.

Dialog box controls

- **Value**: value of the constant; it can be directly typed in the editor or adjusted with the arrows on the right.
- **OK**: button that adds the constant to the equation and closes the dialog box.
- **Cancel**: button that closes the dialog box and cancels the creation of a constant.
- **Help**: button that opens the Trending Tool online help application and provides information on this dialog box.

Section 5.3
Elaborate Statistics on Boolean Variables

What Is in This Section?

This section contains the following topics:

Topic	Page
Variable Statistics	86
Discretisation Step	87
Filter	88
Charts	89

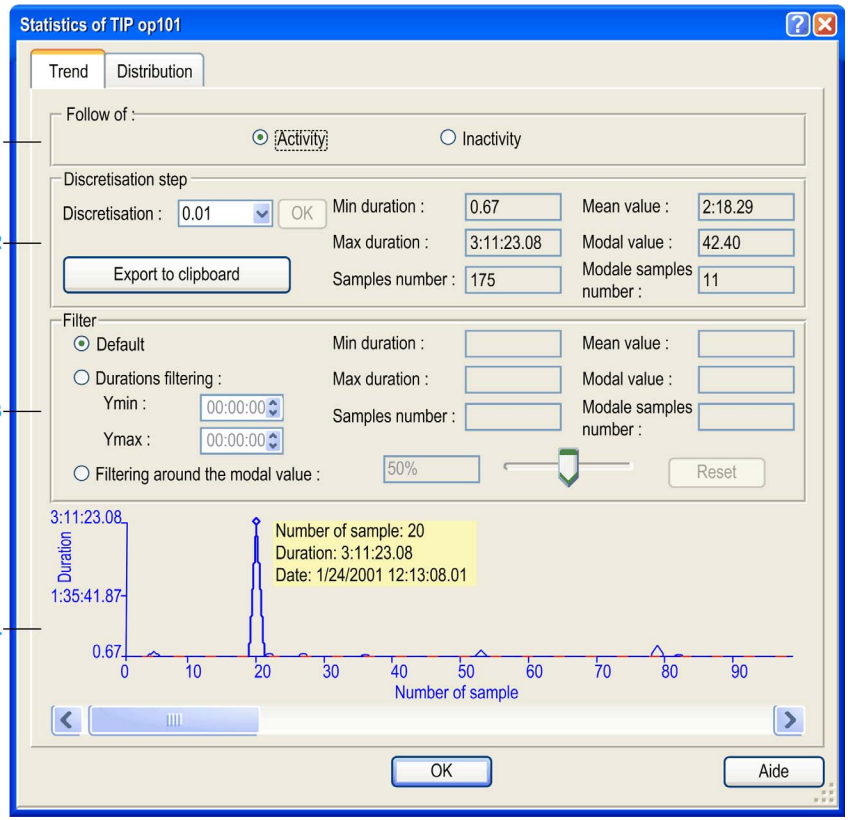
Variable Statistics

At a Glance

When you start analyzing the statistics of a variable, a new window appears. You are then able to perform different adjustments and view two types of graphic, i.e. **Trend** and **Distribution**, by clicking on the tabs.

You can perform different adjustments and close the window. Any time you reopen the statistics of the same variable, your adjustment settings will be restored, provided the .map file has not been closed.

NOTE: Whenever you define a production period in Trending Tool, it is included in the calculation of the statistics of the boolean variable.

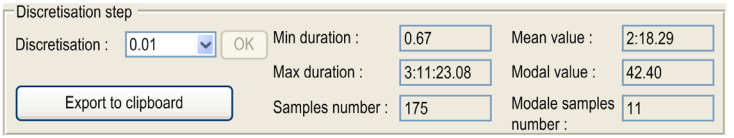


1 Monitor allows you to calculate the statistics based on the variable's activity or inactivity, by clicking the appropriate radio button.
2 Discretisation step
3 Filter
4 Chart

Discretisation Step

At a Glance

The discretisation step enables you to round off the activity/inactivity times according to the required precision. For example: If you have the three following values 6.43 s, 6.45 s, 6.47 s, they become respectively 6.4 s, 6.5 s and 6.5 s with a 0.1 s step.



You can select a predefined step from the drop down menu or type one directly in the drop-down menu area. Typing a discretisation step causes the **OK** button to become active. To confirm what you have typed, click on the **OK** button.

The variable statistic data is:

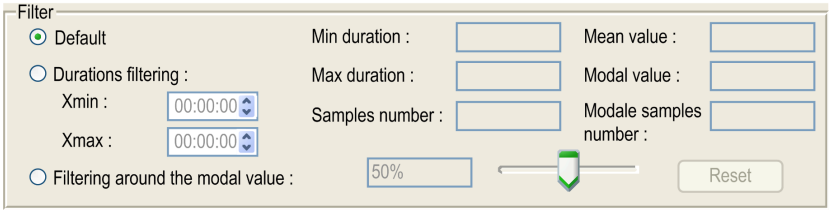
- the minimum activity/inactivity time,
- the maximum activity/inactivity time,
- the number of samples characterizing the variable,
- the average activity/inactivity time,
- the modal value (that is the activity/inactivity time corresponding to the maximum number of occurrences),
- the number of activations/deactivations characterizing the modal value.

You can export the graphic data to the clipboard and operate in Excel by clicking on **Export to the clipboard**.

Filter

At a Glance

The filter enables you to delete the activity/inactivity occurrences of a variable, and then calculate the new statistics.



Default By selecting this option, the calculation of the statistics is performed without the filter (open filter).

Durations filtering When you select this option, you must define a minimum duration and a maximum duration to delete lower duration and upper duration occurrences. A calculation of the new statistical data is performed. This is similar to placing an analysis window on the x-axis for the Distribution curve and on the y-axis for the Trend curve.

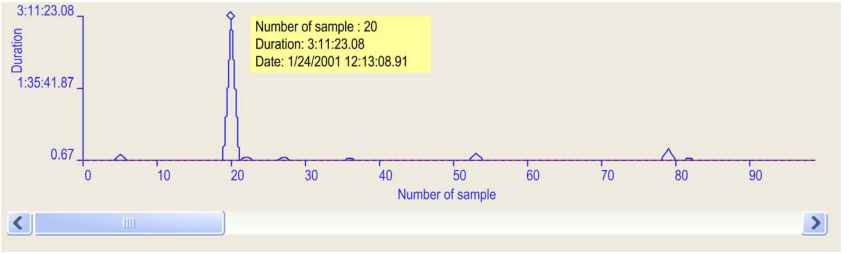
Filtering around the modal value By selecting this option, the minimum and maximum bounds are determined according to a percentage around the duration characterizing the modal value for the whole of the recording. To change these borders, use the slider option.

NOTE: The Reinitialize button enables you to reactivate the minimum duration and maximum duration bounds of the recording if you filter the durations, or reposition the slider on the 50% value around the modal value if you set the borders around the modal value.

Charts

Trend View

You can view two different charts, the trend chart or the distribution chart, according to the tab selected. The trend chart is shown in the figure below.



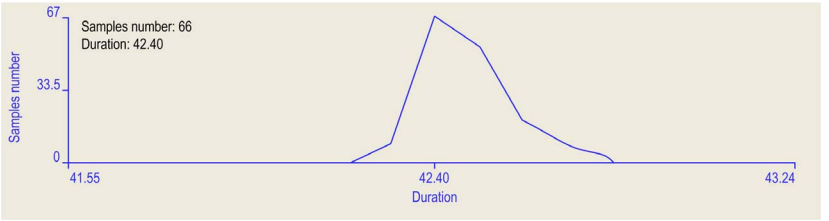
The trend chart represents the variable's activity/inactivity times (on the y-axis) according to the number of successive activity/inactivity occurrences of the variable. For example, the trend chart above shows the development trends of the duration of a mechanical movement while the machine is operating.

You can scroll through the whole chart and when you display the last occurrence level (on the right), in the case of an acquisition in progress, the chart is refreshed as the variable's new developments occur.

You can obtain additional information on an occurrence by moving the mouse pointer onto the chart. An info pop-up will then appear displaying the corresponding occurrence number, activity/inactivity time and date and time.

Distribution View

The distribution chart represents the number of occurrences (activity/inactivity) on the y-axis according to the variable's activity/inactivity times (number of samples for a given duration). For example, the chart below shows the distribution rule of the machine's cycle time.



You can obtain additional information on an occurrence by moving the mouse pointer onto the chart area. The number of activations/deactivations is displayed along with the duration according to where the pointer is located in this area.

Section 5.4
Printing Your Recordings

At a Glance

Trending Tool recordings can be printed just like any other office IT document. This section demonstrates how to print your Trending Tool recordings.

What Is in This Section?

This section contains the following topics:

Topic	Page
Defining a Printing Zone	91
Defining the Page Setup	93
Using the Print Preview	98
Printing	99

Defining a Printing Zone

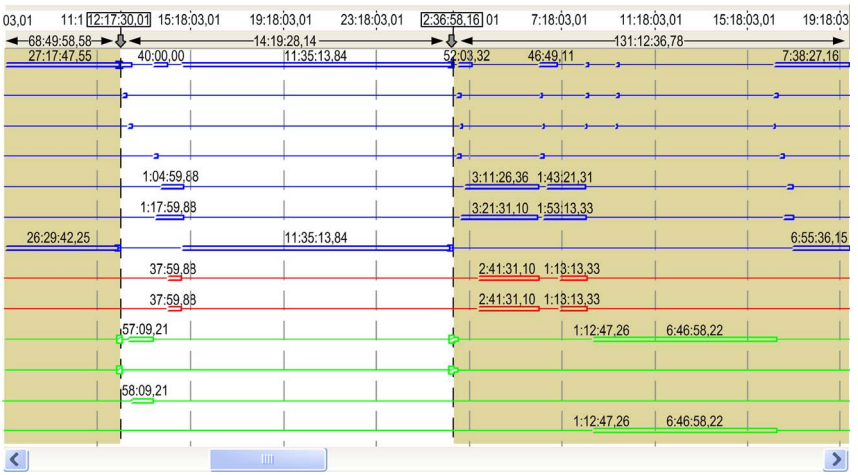
At a Glance

This page explains how to create a printing zone in the module of an open Trending Tool document.

Using the Printing Zone Option

To define the printing zone of the current module of a document, proceed as follows:

Step	Action
1	Stop the recording. Trending Tool can only define a printing zone if the document is closed.
2	Use the **Printing zone** →**Define** command from the **File** menu. This command opens the Trending Tool document selection dialog box.
3	This box contains the list of the time stamps defined in the current module of the document concerned. Select two time stamps to determine the printing zone of your module and click on the **OK** button to close the dialog box and apply the changes made. The white area visible on the trend diagram is the zone of the current module 2 which will be printed. **NOTE:** You can use the **Printing zone** →**Cancel**command from the **File** menu to delete the printing zone defined in the current module of the document concerned. The time stamps will be still be defined in this module.



Using the Printing Dialog Box

The Select dialog box enables you to choose a printing zone, a defined area of your Trending Tool document, you want to print *(see page 90)*.

Dialog Box Options	Description
Printing zone	This zone is determined by a starting stamp and an ending stamp in a list containing the time stamps *(see page 49)* existing in the current module. The time stamp representing the starting point of the printing zone must be selected in the left-hand box; this is called the starting stamp. The time stamp representing the ending point of the printing zone must be selected in the right-hand box; this is called the ending stamp.
OK	Close the dialog box apply the changes made.
Apply	Save the changes made without closing the dialog box.
Cancel	Close the dialog box without applying any changes.

Defining the Page Setup

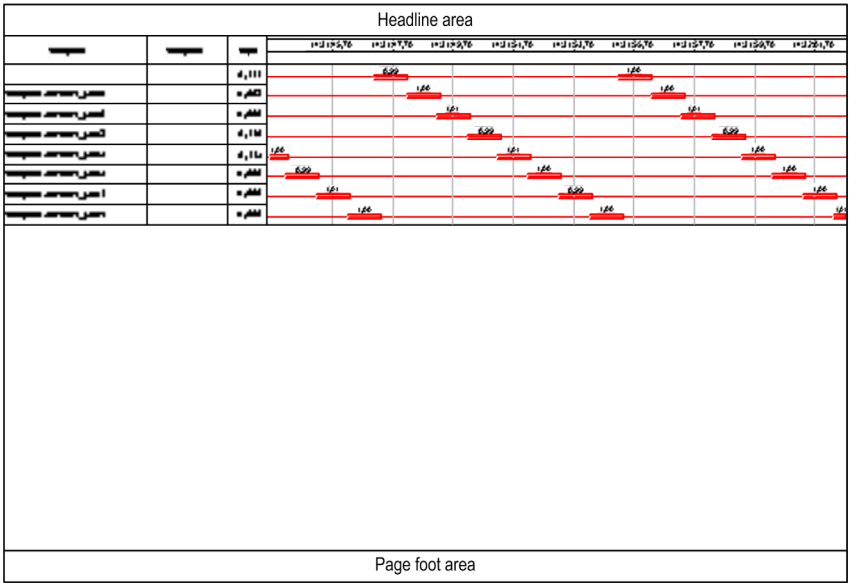
At a Glance

The page setup options found in the dialog box allow you to adjust the diagram of a recording to match the characteristics of the printer, and to the required level of detail.

Accessing the Page Setup Options

To define the page setup of a document, proceed as follows:

Step	Action
1	Stop the recording. Trending Tool can only define a page layout if the document is closed.
2	Use the **Page setup** command from the **File** menu. This command opens the page setup configuration dialog box for Trending Tool documents.
3	Change the Trending Tool page setup properties using the different tabs in the dialog box, then click on the **OK** button to close the dialog box and apply the changes made. **NOTE:** The page setup options available in Trending Tool are detailed in the sections concerning the Trending Tool page setup dialog box.



Using the General Tab Print Option

The Page setup dialog box determines the printing characteristics of a Trending Tool application.

The **General** tab in the Page setup dialog box displays the general page setup parameters. This tab enables you to define the type of document to be printed.

General Tab	
Dialog Box Options	**Description**
Document type	Select the type of document you wish to create in this box. ● Trend diagram: click on this button if you wish to print a trend diagram *(see page 32)*. ● Trend curves: Click on this button if you want to print the trend curves *(see page 46)*. ● Functional analysis: click on this button if you wish to print a functional analysis *(see page 34)*. ● Statistics: click on this button if you want to print statistics *(see page 52)*.
Column printing	If functional analysis is selected, you can choose which column to print. Select the column you wish to print in this box: ● All columns ● From column x to y: Allows you to print columns according to their location in Trending Tool.
OK	Close the dialog box and apply the page setup.
Cancel	Close the dialog box without applying any changes.

Using the Trend Diagram - Trend Curves Tab Print Option

The **Trend Diagram - Trend Curves** tab in the Page setup dialog box displays the parameters related to the margins of the printed document.

This tab enables you to select the data to be printed.

Trend Diagram - Trend Curves Tab	
Dialog Box Options	**Description**
Scale	To print a trend diagram or trend curves, select the scale of the document.
Print modules	Select the modules you wish to print in this box: ● The current module ● All the modules ● All the modules with a printing zone defined ● All the modules according to the current printing zone

Using the Page Tab Print Option

The **Page** tab in the Page setup dialog box displays the parameters related to every page of the document.

This tab presents the layout parameters of the document on the page.

Page Tab	
Dialog Box Options	**Description**
Orientation	This area enables you to choose whether you want to print your document in landscape or portrait mode.
Portrait	Click on this button if you wish to print in portrait mode.
Landscape	Click on this button if you wish to print in landscape mode.
Configuration	This button opens a dialog box in which you can select the printer and its parameters.
Scaling	Use this area to change the size of the printed document.
Scale to	Check this box to change the printing scale of the whole document. Type the required expansion or reduction percentage directly in the editor or use the arrows on the right.
Adjust	Check this box to adjust the size of the printed document to a particular number of pages. Type the required number of pages in the editor or use the arrows on the right.
OK	Close the dialog box and apply the page setup.
Cancel	Close the dialog box without applying any changes.

Using the Margin Tab Print Option

The Margin tab in the Page setup dialog box displays the parameters related to the margins of the printed document.

Margin Tab	
Dialog Box Options	**Description**
Margin	Select the document's margins in this area.
Top	Type in the document's top margin in millimeters in this area.
Left	Type in the document's left margin in millimeters in this area.
Bottom	Type in the document's bottom margin in millimeters in this area.
Right	Type in the document's right margin in millimeters in this area.
Borders	Select a border mode for the printed pages in this area.
On Each Page	Select this option to print a frame on each page of the document.
No Border	Select this option if you do not want to print frames.

Margin Tab	
Dialog Box Options	**Description**
OK	Close the dialog box and apply the page setup.
Cancel	Close the dialog box without applying any changes.

Using the Header/Footer Tab Print Option

The Header and Footer tabs in the Page setup dialog box enable you to define the headers and footers of printed documents.

Trending Tool headers and footers are divided into three areas: left, center and right. The tabs at the top of the box enable you to switch between the areas.

Header/Footer Tab	
Dialog Box Options	**Description**
Left	Select this tab to fill in the left area of the header or footer.
Center	Select this tab to fill in the center area of the header or footer.
Right	Select this tab to fill in the right area of the header or footer.
Text Field	Directly type the text you wish to add to the header or footer in this area. You can add key words that will be reproduced on the printed document, either directly or via different controls beneath the editor.
	Click on this button to change the character font associated with the area. This button opens the standard Windows font management dialog box.
	Click on this button to show the current date in the printed area.
	Click on this button to show the current time in the printed area.
	Click on this button to show the name of the recording file in the printed area.
	Click on this button to show the name of the current module in the printed area.
	Click on this button to show the page number in the printed area.

Header/Footer Tab	
Dialog Box Options	**Description**
	Click on this button to show the number of pages in the document in the printed area.
Add	Select a field from the drop-down list and click on this button to add the field to the printed area.
OK	Close the dialog box and apply the page setup.
Cancel	Close the dialog box without applying any changes.

Using the Options Tab in the Page Setup Dialog Box

Data can be added to or removed from the printed document using the Options tab in the Page setup dialog box.

Options Tab	
Dialog Box Options	**Description**
Print Time Stamps	Check this box to print the recording's time stamps.
Print Alarms	Check this box to print variable activations on the trend diagram's dating scale.
OK	Close the dialog box and apply the page setup.
Cancel	Close the dialog box without applying any changes.

Using the Print Preview

At a Glance

The print preview mode enables you to check the contents of a document before actually printing it.

Activating the Print Preview Mode

To activate the print preview mode, proceed as follows:

Step	Action
1	Stop the recording using the **Stop** button  on the **Recording** tool bar. All the printing functions only work if the Trending Tool document is closed.
2	Use the **Print preview** command from the **Services** menu to run the preview mode. This command opens a window displaying the recording in a format similar to that used by the printer.
3	Scroll through the preview using the buttons on the tool bar located at the top of the window and print the document by clicking on the **Print...** button, or exit this mode by clicking on the **Close** button.

Previewing the Functions of the Tool Bar

The preview functions are all shown on the tool bar. This tool bar contains the following commands:

Print	Open the printing properties dialog box *(see page 99)* to print a document.
Next	Display the next page of the document. This button is gray toned if the last page of the document is displayed.
Previous	Display the following page of the document. This button is gray toned if the first page of the document is displayed.
One page / Two pages	Switch between a one-page display mode and a two-page display mode.
Zoom in	Enlarge the size of the page displayed. The Trending Tool preview features three zoom levels.
Zoom out	Reduce the size of the page displayed.
Close	Close the print preview mode.

Using the Shortcut Keys

In addition to the tool bar buttons, there are shortcut keys that you can use:

- All the pages of the document can be immediately accessed using the scroll bar on the right-hand side of the preview window.
- You can zoom in/out by clicking on the pages of the document.

Printing

At a Glance

Trending Tool uses the standard Windows dialog box for printing documents. For additional information on printing settings, refer to your operating system user documentation.

To print a Trending Tool document, proceed as follows:

Step	Action
1	Stop the recording in progress. Trending Tool can only print closed recordings.
2	Use the **Print...** command from the **Services** menu. This command opens the standard Windows dialog box for defining printing settings.
3	Select the required settings in the dialog box and click on **OK** to print the document.

Section 5.5
Exchanging Data With Another Application

Copy, Paste, and Drag and Drop

At a Glance

This section explains how to transfer data from Trending Tool to other applications by using the copy and paste or the drag and drop functions.

The copy and paste or drag and drop functions enable you to set up a configuration in Trending Tool quickly. You can also use these functions in the data tables, FA diagram list and statistical data table, to easily transfer data to other applications.

To copy the statistical data table of a module to Excel, proceed as follows:

Step	Action
1	Display the data table of the chosen module.
2	Use the **Select all...** command from the **Edit** menu to select all of the module's variables. The color of the lines in the table will change, indicating that they have been selected.
3	Use the **Copy** command from the **Edit** menu to copy the statistical data table to the clipboard.
4	Open the Excel application, click on the cell where you want to locate the top left-hand corner of the data table and use the **Paste** command from the **Edit** menu in this application. The data table is fully pasted into Excel.

This procedure can be adapted if you want to copy and paste part of the data table information or the functional analysis lines.

NOTE: Do not attempt to transfer quantities of data that are incompatible with the capacity of the destination applications. For example, you cannot create Excel tables of more than 65,000 lines.

Section 5.6
Managing your Recordings

Creating & Saving Your Recordings

Saving a Recording by Manual Backup

Once the recording is finished, you must save the data onto your computer as it is not automatically stored by Trending Tool. To save a configuration or a recording:

Step	Action
1	Use the **Save trend** command from the **Services** menu. If this is the first time that the document has been saved, Trending Tool will open the standard Windows **Save as** dialog box.
2	In the dialog box, type a name for the recording and select the location where you want to save the file. Click on the **OK** button to close the dialog box and save the document.

Saving a Recording Using Autosave

NOTE: This procedure can only be executed during acquisition.

To protect yourself from system failures during data acquisition, you can perform recurring and automatic backups of the document.

Step	Action
1	The document will have to be saved manually at least once in order to link it to a file.
2	Before starting acquisition, configure the save period time. Click on the Recording options command from the Services menu to open the Recording options dialog box. In the General tab, activate and define the Save period and confirm. Autosave is now enabled for any acquisition from a previously saved document.
3	Once the recording is finished, you must save the document one final time to take account of the developments acquired since the last autosave.

Saving a Recording Under Another Name

To save a recording under another name or to a different location:

Step	Action
1	Use the **Save trend** command from the **Services** menu. This command opens the standard Windows **Save as** dialog box.
2	In the dialog box, type a name for the recording and select the location where you want to save the file. Click on the **OK** button to close the dialog box and save the document.

Opening a Previously Saved Recording

To open a previously saved recording or configuration in Trending Tool:

Step	Action
1	Use the **Open trend** command from the **Services** menu. This command opens the standard Windows **Open** dialog box.
2	In the dialog box, select the required file and click on the **OK** button to open a Trending Tool recording window containing the selected recording.

Starting a New Recording

To start a new recording from an existing configuration:

Step	Action
1	Open an Trending Tool file *(see page 102)* containing the configuration to apply to a new recording.
2	Use the **Start recording** button from the Trending tool bar to start recording developments.
3	Trending Tool opens a dialog box asking you what you want to do with the previous recording. Answer that you want to keep the previously acquired developments.
4	Save the new recording.

Starting a New Recording using the Autosave

To start a new recording from an existing configuration:

Step	Action
1	Open an Trending Tool file *(see page 102)* containing the configuration to apply to a new recording.
2	Click on the **Recording options** command from the **Services** menu to open the **Recording options** dialog box.

Step	Action
3	In the **General** tab, activate and define the **Save period** and confirm. Autosave is now enabled for any acquisition from a previously saved document.
4	Save the file under a new name.
5	Use the **Start recording** button from the Trending tool bar to start recording developments.
6	Trending Tool opens a dialog box asking you what you want to do with the previous recording. Answer that you want to overwrite the previously acquired developments.

Managing Large Files

Trending Tool recordings must be managed in such a way that you can obtain a recording time suited to statistical calculations without overloading the computer. The software does not limit the size of the processed documents. A recording may contain a large number of variables that can be monitored for more than 30 days. The size of the recording depends directly on these two criteria as well as on the rate of the monitored equipment.

However, it is not easy to estimate the size at which a file becomes to large to be processed correctly by Trending Tool. The performance of the computer plays an important part. Tests carried out by Schneider Electric show that, for a recording size of 20 MB, the average processing time for a user interface operation is between 15 and 20 seconds on a Pentium II computer running at 300 MHz with 64 MB of RAM.

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